

EXAMPLE 2 | Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its linearization there. Then use it to approximate $f(1.1, -0.1)$.

SOLUTION The partial derivatives are

$$\begin{aligned}f_x(x, y) &= e^{xy} + xye^{xy} & f_y(x, y) &= x^2e^{xy} \\f_x(1, 0) &= 1 & f_y(1, 0) &= 1\end{aligned}$$

Figure 5 shows the graphs of the function f and its linearization L in Example 2.

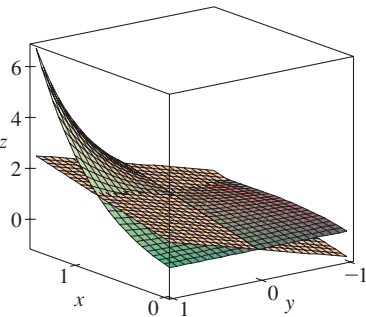


FIGURE 5

Both f_x and f_y are continuous functions, so f is differentiable by Theorem 6. The linearization is

$$\begin{aligned}L(x, y) &= f(1, 0) + f_x(1, 0)(x - 1) + f_y(1, 0)(y - 0) \\&= 1 + 1(x - 1) + 1 \cdot y = x + y\end{aligned}$$

The corresponding linear approximation is

$$xe^{xy} \approx x + y$$

so
$$f(1.1, -0.1) \approx 1.1 - 0.1 = 1$$

Compare this with the actual value of $f(1.1, -0.1) = 1.1e^{-0.11} \approx 0.98542$. ■

EXAMPLE 3 | **Heat index** At the beginning of Section 9.2 we discussed the heat index (perceived temperature) I as a function of the actual temperature T and the relative humidity H and gave the following table of values from the National Weather Service.

		Relative humidity (%)								
Actual temperature (°F)	$T \backslash H$	50	55	60	65	70	75	80	85	90
	90	96	98	100	103	106	109	112	115	119
	92	100	103	105	108	112	115	119	123	128
	94	104	107	111	114	118	122	127	132	137
	96	109	113	116	121	125	130	135	141	146
	98	114	118	123	127	133	138	144	150	157
	100	119	124	129	135	141	147	154	161	168

Find a linear approximation for the heat index $I = f(T, H)$ when T is near 96°F and H is near 70% . Use it to estimate the heat index when the temperature is 97°F and the relative humidity is 72% .

SOLUTION We read from the table that $f(96, 70) = 125$. In Section 9.2 we used the tabular values to estimate that $f_T(96, 70) \approx 3.75$ and $f_H(96, 70) \approx 0.9$. (See pages 585–86.) So the linear approximation is

$$\begin{aligned}f(T, H) &\approx f(96, 70) + f_T(96, 70)(T - 96) + f_H(96, 70)(H - 70) \\&\approx 125 + 3.75(T - 96) + 0.9(H - 70)\end{aligned}$$

In particular,

$$f(97, 72) \approx 125 + 3.75(1) + 0.9(2) = 130.55$$

Therefore, when $T = 97^\circ\text{F}$ and $H = 72\%$, the heat index is

$$I \approx 131^\circ\text{F}$$

EXAMPLE 4 | The wind-chill index $W(T, v)$ is the subjective temperature that we perceive when the actual temperature is T and the wind speed is v . (See Example 9.1.1.) If T is measured in $^\circ\text{C}$ and v in km/h , then a model for W is

$$W(T, v) = 13.12 + 0.6215T - 11.37v^{0.16} + 0.3965Tv^{0.16}$$

Use this equation to find the linearization of W at $(-12, 20)$.

SOLUTION The partial derivatives of W are

$$W_T(T, v) = 0.6215 + 0.3965v^{0.16}$$

$$W_v(T, v) = -1.8192v^{-0.84} + 0.06344Tv^{-0.84}$$

Evaluating W , W_T , and W_v to two decimal places when $T = -12$ and $v = 20$, we get

$$W(-12, 20) = -20.38 \quad W_T(-12, 20) = 1.26 \quad W_v(-12, 20) = -0.21$$

So the linearization of the wind chill function at $(-12, 20)$ is

$$L(T, v) = -20.38 + 1.26(T + 12) - 0.21(v - 20)$$

This linear function is a good approximation to the wind chill function $W(T, v)$ when T is near -12°C and v is near 20 km/h . The accuracy of this approximation decreases as we move away from these values.

Linear approximations and differentiability can be defined in a similar manner for functions of more than two variables. A differentiable function is defined by an expression similar to the one in Definition 5. For functions of three variables the **linear approximation** is

$$f(x, y, z) \approx f(a, b, c) + f_x(a, b, c)(x - a) + f_y(a, b, c)(y - b) + f_z(a, b, c)(z - c)$$

and the linearization $L(x, y, z)$ is the right side of this expression.

EXERCISES 9.3

1–6 Find an equation of the tangent plane to the given surface at the specified point.

1. $z = 3y^2 - 2x^2 + x$, $(2, -1, -3)$

2. $z = 3(x - 1)^2 + 2(y + 3)^2 + 7$, $(2, -2, 12)$

3. $z = \sqrt{xy}$, $(1, 1, 1)$

4. $z = xe^{xy}$, $(2, 0, 2)$

5. $z = y \cos(x - y)$, $(2, 2, 2)$

6. $z = \ln(x - 2y)$, $(3, 1, 0)$

7–12 Explain why the function is differentiable at the given point. Then find the linearization of the function at that point.

7. $f(x, y) = x\sqrt{y}$, $(1, 4)$

8. $f(x, y) = x^3y^4$, $(1, 1)$

9. $f(x, y) = \frac{x}{x + y}, \quad (2, 1)$
10. $f(x, y) = \sqrt{x + e^{4y}}, \quad (3, 0)$
11. $f(x, y, z) = x^2y + y^2z, \quad (1, 2, 3)$
12. $f(x, y, z) = e^{-xy} \cos z, \quad (0, 2, 0)$

13–14 Verify the linear approximation at $(0, 0)$.

13. $\frac{2x + 3}{4y + 1} \approx 3 + 2x - 12y$ 14. $\sqrt{y + \cos^2 x} \approx 1 + \frac{1}{2}y$

15. Given that f is a differentiable function with $f(2, 5) = 6$, $f_x(2, 5) = 1$, and $f_y(2, 5) = -1$, use a linear approximation to estimate $f(2.2, 4.9)$.

16. Find the linear approximation of the function

$$f(x, y) = \ln(x - 3y)$$

at $(7, 2)$ and use it to approximate $f(6.9, 2.06)$.

17. Find the linear approximation of the function

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

at $(3, 2, 6)$ and use it to approximate the number

$$\sqrt{(3.02)^2 + (1.97)^2 + (5.99)^2}$$

18. The wave heights h in the open sea depend on the speed v of the wind and the length of time t that the wind has been blowing at that speed. Values of the function $h = f(v, t)$ are recorded in feet in the following table. Use the table to find a linear approximation to the wave height function when v is near 40 knots and t is near 20 hours. Then estimate the wave heights when the wind has been blowing for 24 hours at 43 knots.

		Duration (hours)						
Wind speed (knots)	$v \backslash t$	5	10	15	20	30	40	50
	20	5	7	8	8	9	9	9
	30	9	13	16	17	18	19	19
	40	14	21	25	28	31	33	33
	50	19	29	36	40	45	48	50
	60	24	37	47	54	62	67	69

19. Use the table in Example 3 to find a linear approximation to the heat index function when the temperature is near 94°F and the relative humidity is near 80%. Then estimate the heat index when the temperature is 95°F and the relative humidity is 78%.

20. The **wind-chill index** W is the perceived temperature when the actual temperature is T and the wind speed is v , so we can write $W = f(T, v)$. The following table of values is an excerpt from Table 9.1.1. Use the table to find a linear approximation to the wind-chill index function when T is near -15°C and v is near 50 km/h. Then estimate the wind-chill index when the temperature is -17°C and the wind speed is 55 km/hr.

		Wind speed (km/h)					
Actual temperature (°C)	$T \backslash v$	20	30	40	50	60	70
	-10	-18	-20	-21	-22	-23	-23
	-15	-24	-26	-27	-29	-30	-30
	-20	-30	-33	-34	-35	-36	-37
	-25	-37	-39	-41	-42	-43	-44

21. **Lizard energy expenditure** The average energy E (in kcal) needed for a lizard to walk or run a distance of 1 km has been modeled by the equation

$$E(m, v) = 2.65m^{0.66} + \frac{3.5m^{0.75}}{v}$$

where m is the body mass of the lizard (in grams) and v is its speed (in km/h). Find the linearization of the energy function at $(400, 8)$.

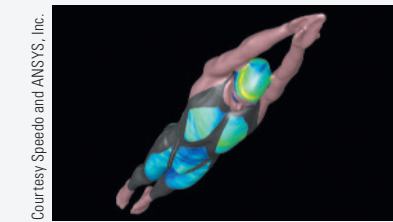
Source: Adapted from C. Robbins, *Wildlife Feeding and Nutrition*, 2nd ed. (San Diego: Academic Press, 1993).

22. **Body surface area** A model for the surface area of a human body is given by the function

$$S = f(w, h) = 0.1091w^{0.425}h^{0.725}$$

- where w is the weight (in pounds), h is the height (in inches), and S is measured in square feet.
- (a) Calculate the linearization $L(w, h)$ of this function when $w = 160$ lb and $h = 70$ in.
- (b) Compare the values of $f(w, h)$ and $L(w, h)$ when $(w, h) = (162, 73)$ and when $(w, h) = (170, 80)$. Comment on the relative accuracy of the approximations.

PROJECT The Speedo LZR Racer



Many technological advances have occurred in sports that have contributed to increased athletic performance. One of the best known is the introduction, in 2008, of the Speedo LZR racer. It was claimed that this full-body swimsuit reduced a swimmer's drag in the water. Figure 1 shows the number of world records broken in men's and women's long-course freestyle swimming events from 1990 to 2011.¹ The dramatic increase in 2008 when the suit was introduced led people to claim that such suits are a form of technological doping. As a result all full-body suits were banned from competition starting in 2010.

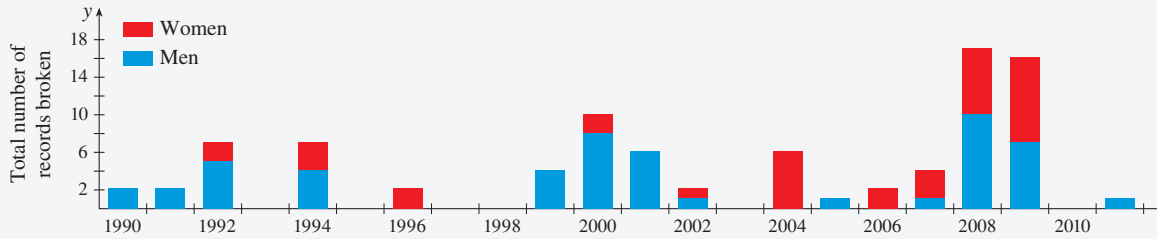


FIGURE 1 Number of world records set in long-course men's and women's freestyle swimming event 1990–2011

It might be surprising that a simple reduction in drag could have such a big effect on performance. We can gain some insight into this using a simple mathematical model.²

The speed v of an object being propelled through water is given by

$$v(P, C) = \left(\frac{2P}{kC} \right)^{1/3}$$

where P is the power being used to propel the object, C is the drag coefficient, and k is a positive constant. Athletes can therefore increase their swimming speeds by increasing their power or reducing their drag coefficients. But how effective is each of these?

To compare the effect of increasing power versus reducing drag, we need to somehow compare the two in common units. The most common approach is to determine the *percentage change* in speed that results from a given *percentage change* in power and in drag.

If we work with percentages as fractions, then when power is changed by a fraction x (with x corresponding to $100x$ percent), P changes from P to $P + xP$. Likewise, if the drag coefficient is changed by a fraction y , this means that it has changed from C to $C + yC$. Finally, the fractional change in speed resulting from both effects is

$$(1) \quad \frac{v(P + xP, C + yC) - v(P, C)}{v(P, C)}$$

1. Expression 1 gives the fractional change in speed that results from a change x in power and a change y in drag. Show that this reduces to the function

$$f(x, y) = \left(\frac{1 + x}{1 + y} \right)^{1/3} - 1$$

Given the context, what is the domain of f ?

1. L. Foster et al., "Influence of Full Body Swimsuits on Competitive Performance," *Procedia Engineering* 34 (2012): 712–17.

2. Adapted from <http://plus.maths.org/content/swimming>.

2. Suppose that the possible changes in power x and drag y are small. Find the linear approximation to the function $f(x, y)$. What does this approximation tell you about the effect of a small increase in power versus a small decrease in drag?
3. Calculate $f_{xx}(x, y)$ and $f_{yy}(x, y)$. Based on the signs of these derivatives, does the linear approximation in Problem 2 result in an overestimate or an underestimate for an increase in power? What about for a decrease in drag? Use your answer to explain why, for changes in power or drag that are not very small, a decrease in drag is more effective.
4. Graph the level curves of $f(x, y)$. Explain how the shapes of these curves relate to your answers to Problems 2 and 3.

9.4 The Chain Rule

Recall that the Chain Rule for functions of a single variable gives the rule for differentiating a composite function: If $y = f(x)$ and $x = g(t)$, where f and g are differentiable functions, then y is indirectly a differentiable function of t and

$$(1) \quad \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$$

For functions of two variables we have $z = f(x, y)$, where each of the variables x and y is a function of a variable t . This means that z is indirectly a function of t , $z = f(g(t), h(t))$, and the Chain Rule gives a formula for differentiating z as a function of t . We assume that f is differentiable (Definition 9.3.5). Recall that this is the case when f_x and f_y are continuous (Theorem 9.3.6).

(2) The Chain Rule Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

A rigorous proof of the Chain Rule is rather technical and is similar to the proof of the single-variable case at the end of Section 3.5. Instead, we give a more intuitive indication of why it is true.

Near a point (a, b) in the domain of f , we approximate the differentiable function f by its linear approximation:

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Let $\Delta x = x - a$, $\Delta y = y - b$, and $\Delta z = f(x, y) - f(a, b)$. Then we can write

$$\Delta z \approx \frac{\partial z}{\partial x} \Delta x + \frac{\partial z}{\partial y} \Delta y$$

A change of Δt in t produces changes of Δx in x and Δy in y which, in turn, produce a

change of Δz in z . Dividing both sides of this last equation by Δt , we get

$$\frac{\Delta z}{\Delta t} \approx \frac{\partial z}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial z}{\partial y} \frac{\Delta y}{\Delta t}$$

If we now let $\Delta t \rightarrow 0$, this approximation becomes better and we have

$$\frac{\Delta z}{\Delta t} \rightarrow \frac{dz}{dt} \quad \frac{\Delta x}{\Delta t} \rightarrow \frac{dx}{dt} \quad \frac{\Delta y}{\Delta t} \rightarrow \frac{dy}{dt}$$

So it seems reasonable that

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

EXAMPLE 1 | If $z = x^2y + 3xy^4$, where $x = \sin 2t$ and $y = \cos t$, find dz/dt when $t = 0$.

SOLUTION The Chain Rule gives

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \\ &= (2xy + 3y^4)(2 \cos 2t) + (x^2 + 12xy^3)(-\sin t) \end{aligned}$$

It's not necessary to substitute the expressions for x and y in terms of t . We simply observe that when $t = 0$, we have $x = \sin 0 = 0$ and $y = \cos 0 = 1$. Therefore

$$\left. \frac{dz}{dt} \right|_{t=0} = (0 + 3)(2 \cos 0) + (0 + 0)(-\sin 0) = 6$$

Strictly speaking, it's not absolutely necessary to use Theorem 2 to solve Example 1. Instead we could have substituted $x = \sin 2t$ and $y = \cos t$ into the expression for z and then used the ordinary chain rule. But in the next example it really is necessary to use Theorem 2.



EXAMPLE 2 | **BB** Tuna biomass depends on the availability of the small fish that they eat (such as puffer fish and trigger fish) as well as the size of the catch of the annual commercial tuna fishery (currently about four million tons per year). Use the Chain Rule to discuss whether the tuna biomass is increasing or decreasing.

SOLUTION Let T be the tuna biomass and S the biomass of the small fish that tuna eat. Let C be the size of the annual tuna catch. Then we can write $T = f(S, C)$. If t denotes time, then the Chain Rule says that

$$(3) \quad \frac{dT}{dt} = \frac{\partial T}{\partial S} \frac{dS}{dt} + \frac{\partial T}{\partial C} \frac{dC}{dt}$$

If we suppose that the population of small fish is increasing, then dS/dt is positive. What about $\partial T/\partial S$? If S increases and C remains the same, then the tuna have more to eat and so T also increases. Thus $\partial T/\partial S > 0$. So the first term on the right side of (3) is positive.

If S remains the same and the catch increases, then T will decrease, which means that $\partial T/\partial C$ is negative. What is the sign of dC/dt ? The size of the tuna catch has been increasing in recent decades but overfishing has resulted in international quotas, which

are sometimes ignored. If C continues to increase, then $dC/dt > 0$ and so the second term in (3) is negative. From Equation 3 we see that dT/dt is the sum of a positive term and a negative term, so we are unable to determine whether dT/dt is positive or negative. On the other hand, if the catch decreases, then the second term is positive and so dT/dt is positive and the tuna biomass increases. ■

EXAMPLE 3 | The pressure P (in kilopascals), volume V (in liters), and temperature T (in kelvins) of a mole of an ideal gas are related by the equation $PV = 8.31T$. Find the rate at which the pressure is changing when the temperature is 300 K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

SOLUTION If t represents the time elapsed in seconds, then at the given instant we have $T = 300$, $dT/dt = 0.1$, $V = 100$, $dV/dt = 0.2$. Since

$$P = 8.31 \frac{T}{V}$$

the Chain Rule gives

$$\begin{aligned} \frac{dP}{dt} &= \frac{\partial P}{\partial T} \frac{dT}{dt} + \frac{\partial P}{\partial V} \frac{dV}{dt} = \frac{8.31}{V} \frac{dT}{dt} - \frac{8.31T}{V^2} \frac{dV}{dt} \\ &= \frac{8.31}{100} (0.1) - \frac{8.31(300)}{100^2} (0.2) = -0.04155 \end{aligned}$$

The pressure is decreasing at a rate of about 0.042 kPa/s. ■

For functions of three variables, where $w = f(x, y, z)$ is differentiable and x , y , and z are differentiable functions of t , the Chain Rule has an extra term:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

EXAMPLE 4 | Find dw/dt if $w = xe^{y/z}$, where $x = 3t + 2$, $y = t^2$, and $z = t^3 - 1$.

SOLUTION By the Chain Rule, we have

$$\begin{aligned} \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \\ &= e^{y/z} \cdot 3 + xe^{y/z} \cdot \frac{1}{z} \cdot 2t + xe^{y/z} \left(-\frac{y}{z^2} \right) \cdot 3t^2 \\ &= e^{y/z} \left(3 + \frac{2xt}{z} - \frac{3xyt^2}{z^2} \right) \end{aligned}$$

■ Implicit Differentiation

The Chain Rule can be used to give a more complete description of the process of implicit differentiation that was introduced in Sections 3.5 and 9.2. We suppose that an equation of the form $F(x, y) = 0$ defines y implicitly as a differentiable function of x , that is, $y = f(x)$, where $F(x, f(x)) = 0$ for all x in the domain of f . If F is differentiable, we can apply the Chain Rule to differentiate both sides of the equation $F(x, y) = 0$ with

respect to x . Since both x and y are functions of x , we obtain

$$\frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0$$

But $dx/dx = 1$, so if $\partial F/\partial y \neq 0$ we solve for dy/dx and obtain

(4)

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}$$

To derive this equation we assumed that $F(x, y) = 0$ defines y implicitly as a function of x . The **Implicit Function Theorem**, proved in advanced calculus, gives conditions under which this assumption is valid: It states that if F is defined on a disk containing (a, b) , where $F(a, b) = 0$, $F_y(a, b) \neq 0$, and F_x and F_y are continuous on the disk, then the equation $F(x, y) = 0$ defines y as a function of x near the point (a, b) and the derivative of this function is given by Equation 4.

EXAMPLE 5 | Find y' if $x^3 + y^3 = 6xy$.

SOLUTION The given equation can be written as

$$F(x, y) = x^3 + y^3 - 6xy = 0$$

so Equation 4 gives

The solution to Example 5 should be compared to the one in Example 3.5.12.

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{3x^2 - 6y}{3y^2 - 6x} = -\frac{x^2 - 2y}{y^2 - 2x}$$

Now we suppose that z is given implicitly as a function $z = f(x, y)$ by an equation of the form $F(x, y, z) = 0$. This means that $F(x, y, f(x, y)) = 0$ for all (x, y) in the domain of f . If F and f are differentiable, then we can use the Chain Rule to differentiate the equation $F(x, y, z) = 0$ as follows:

$$\frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$$

But

$$\frac{\partial}{\partial x}(x) = 1 \quad \text{and} \quad \frac{\partial}{\partial x}(y) = 0$$

so this equation becomes

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0$$

If $\partial F/\partial z \neq 0$, we solve for $\partial z/\partial x$ and obtain the first formula in Equations 5. The formula for $\partial z/\partial y$ is obtained in a similar manner.

(5)

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

Again, a version of the **Implicit Function Theorem** stipulates conditions under which our assumption is valid: If F is defined within a sphere containing (a, b, c) , where $F(a, b, c) = 0$, $F_z(a, b, c) \neq 0$, and F_x , F_y , and F_z are continuous inside the sphere, then the equation $F(x, y, z) = 0$ defines z as a function of x and y near the point (a, b, c) and this function is differentiable, with partial derivatives given by (5).

EXAMPLE 6 | BB Infectious disease outbreak size Epidemiologists often wish to predict the fraction of the population that will ultimately be infected when a disease begins to spread. Mathematical models have been used to do so. The Kermack-McKendrick model leads to the following equation (see the project on page 354):

$$\rho e^{-qA} = 1 - A$$

where A is the fraction of the population ultimately infected, q is a measure of disease transmissibility, and ρ is a measure of the fraction of the population that is initially susceptible to infection. How does the outbreak size A change with an increase in the transmissibility q ?

SOLUTION Let $F(\rho, q, A) = \rho e^{-qA} - 1 + A$. Then, from Equations 5, the rate of change of A with respect to q is

$$\frac{\partial A}{\partial q} = -\frac{F_q}{F_A} = -\frac{-\rho A e^{-qA}}{-\rho q e^{-qA} + 1} = \frac{\rho A}{e^{qA} - \rho q}$$

This is the rate of increase of the outbreak size as the transmissibility increases while ρ remains constant. ■

EXERCISES 9.4

1–6 Use the Chain Rule to find dz/dt or dw/dt .

- $z = x^2 + y^2 + xy$, $x = \sin t$, $y = e^t$
- $z = \cos(x + 4y)$, $x = 5t^4$, $y = 1/t$
- $z = \sqrt{1 + x^2 + y^2}$, $x = \ln t$, $y = \cos t$
- $z = \tan^{-1}(y/x)$, $x = e^t$, $y = 1 - e^{-t}$
- $w = x e^{y/z}$, $x = t^2$, $y = 1 - t$, $z = 1 + 2t$
- $w = \ln \sqrt{x^2 + y^2 + z^2}$, $x = \sin t$, $y = \cos t$, $z = \tan t$

7. Suppose $z = f(x, y)$, where $x = g(s, t)$ and $y = h(s, t)$ are differentiable functions of s and t . Use Theorem 2 to show that

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

8–10 Use Exercise 7 to find $\partial z/\partial s$ and $\partial z/\partial t$.

- $z = e^x \sin y$, $x = st^2$, $y = s^2t$
- $z = x^2y^3$, $x = s \cos t$, $y = s \sin t$

10. $z = e^{x+2y}$, $x = s/t$, $y = t/s$

11. If $z = f(x, y)$, where f is differentiable, and

$$x = g(t) \quad y = h(t)$$

$$g(3) = 2 \quad h(3) = 7$$

$$g'(3) = 5 \quad h'(3) = -4$$

$$f_s(2, 7) = 6 \quad f_t(2, 7) = -8$$

find dz/dt when $t = 3$.

12. Let $W(s, t) = F(u(s, t), v(s, t))$, where F , u , and v are differentiable, and

$$u(1, 0) = 2 \quad v(1, 0) = 3$$

$$u_s(1, 0) = -2 \quad v_s(1, 0) = 5$$

$$u_t(1, 0) = 6 \quad v_t(1, 0) = 4$$

$$F_u(2, 3) = -1 \quad F_v(2, 3) = 10$$

Find $W_s(1, 0)$ and $W_t(1, 0)$.

13–16 Use Equation 4 to find dy/dx .

13. $y \cos x = x^2 + y^2$

14. $y^5 + x^2y^3 = 1 + ye^{x^2}$

15. $\cos(x - y) = xe^y$

16. $\sin x + \cos y = \sin x \cos y$

17–20 Use Equations 5 to find $\partial z/\partial x$ and $\partial z/\partial y$.

17. $x^2 + y^2 + z^2 = 3xyz$

18. $xyz = \cos(x + y + z)$

19. $x - z = \arctan(yz)$

20. $yz = \ln(x + z)$

- 21. Gray wolves** range over Canada, the northern United States, Europe, and Asia. They feed on animals such as moose, deer, elk, and caribou and smaller animals such as hare and lynx. Their competitors are principally coyotes and bears. They have few predators, aside from humans. If W denotes the size of the gray wolf population, we can write

$$W = f(F, C)$$

where F is the size of the food supply and C is the number of competitors.

- The variables W , F , and C are all functions of time t . Write out the Chain Rule for this situation.
- Are the partial derivatives $\partial W/\partial F$ and $\partial W/\partial C$ positive or negative?
- What happens if the food supply increases while the competition decreases?
- What can you say if both F and C increase?



- 22. Wheat production** W in a given year depends on the average temperature T and the annual rainfall R . Scientists estimate that the average temperature is rising at a rate of $0.15^\circ\text{C}/\text{year}$ and rainfall is decreasing at a rate of $0.1 \text{ cm}/\text{year}$. They also estimate that, at current production levels, $\partial W/\partial T = -2$ and $\partial W/\partial R = 8$.
- What is the significance of the signs of these partial derivatives?
 - Estimate the current rate of change of wheat production, dW/dt .

- 23. Body mass index** Recall from Example 9.1.2 that the BMI of a person is

$$B(m, h) = \frac{m}{h^2}$$

where m is the person's mass (in kilograms) and h is the height (in meters). Both m and h depend on the person's age a . Use the Chain Rule to find an expression for the rate of change of the BMI with respect to age.

- 24. Infectious disease control** In Example 9.1.12 we looked at the following model for the spread of SARS:

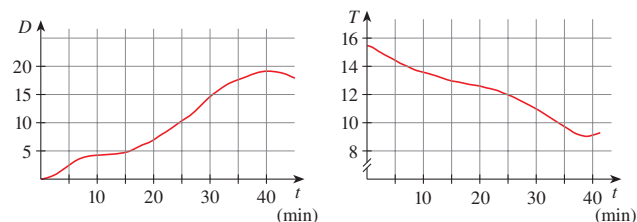
$$R_0(d, v) = 5(1 - v) \frac{d}{1 + d}$$

where R_0 is the basic reproduction number, v is the fraction of the population that is vaccinated, and d is the average number of days that an infected individual remains in the population before being quarantined. The quantities d and v depend on the investment I that the government makes to contain the disease. Use the Chain Rule to obtain an expression for the rate of change of R_0 with respect to investment.

- 25. Infectious disease outbreak size** In Example 6 we used the equation $\rho e^{-qA} = 1 - A$ to model the relationship among A (the fraction of the population ultimately infected by the disease), q (a measure of disease transmissibility), and ρ (a measure of the fraction of the population that is initially susceptible to infection). How does the outbreak size A change with an increase in ρ ?
- 26.** The speed of sound traveling through ocean water with salinity 35 parts per thousand has been modeled by the equation

$$C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 + 0.016D$$

where C is the speed of sound (in meters per second), T is the temperature (in degrees Celsius), and D is the depth below the ocean surface (in meters). A scuba diver began a leisurely dive into the ocean water; the diver's depth and the surrounding water temperature over time are recorded in the following graphs. Estimate the rate of change (with respect to time) of the speed of sound through the ocean water experienced by the diver 20 minutes into the dive. What are the units?



- 27.** The pressure of 1 mole of an ideal gas is increasing at a rate of 0.05 kPa/s and the temperature is increasing at a rate of 0.15 K/s . Use the equation in Example 3 to find the rate of change of the volume when the pressure is 20 kPa and the temperature is 320 K .
- 28.** If a sound with frequency f_s is produced by a source traveling along a line with speed v_s and an observer is

traveling with speed v_o along the same line from the opposite direction toward the source, then the frequency of the sound heard by the observer is

$$f_o = \left(\frac{c + v_o}{c - v_s} \right) f_s$$

where c is the speed of sound, about 332 m/s. (This is the *Doppler effect*.) Suppose that, at a particular moment, you are in a train traveling at 34 m/s and accelerating at 1.2 m/s^2 . A train is approaching you from the opposite direction on the other track at 40 m/s, accelerating at 1.4 m/s^2 , and sounds its whistle, which has a frequency of 460 Hz. At that instant, what is the perceived frequency that you hear and how fast is it changing?

29. Suppose that the equation $F(x, y, z) = 0$ implicitly defines each of the three variables x , y , and z as functions of the other two: $z = f(x, y)$, $y = g(x, z)$, $x = h(y, z)$. If F is differentiable and F_x , F_y , and F_z are all nonzero, show that

$$\frac{\partial z}{\partial x} \frac{\partial x}{\partial y} \frac{\partial y}{\partial z} = -1$$

30. Equation 4 is a formula for the derivative dy/dx of a function defined implicitly by an equation $F(x, y) = 0$, provided that F is differentiable and $F_y \neq 0$. Prove that if F has continuous second derivatives, then a formula for the second derivative of y is

$$\frac{d^2y}{dx^2} = -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3}$$

9.5 Directional Derivatives and the Gradient Vector

The weather map in Figure 1 shows a contour map of the temperature function $T(x, y)$ for the states of California and Nevada at 3:00 PM on a day in October. The level curves, or isothermals, join locations with the same temperature. The partial derivative T_x at a location such as Reno is the rate of change of temperature with respect to distance if we travel east from Reno; T_y is the rate of change of temperature if we travel north. But what if we want to know the rate of change of temperature when we travel southeast (toward Las Vegas), or in some other direction? In this section we introduce a type of derivative, called a *directional derivative*, that enables us to find the rate of change of a function of two variables in any direction.

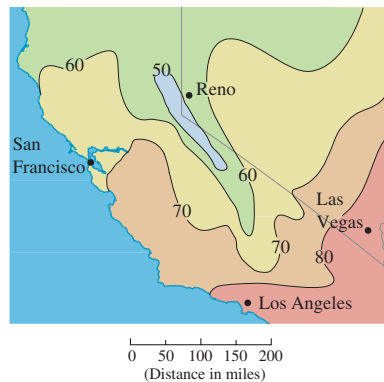
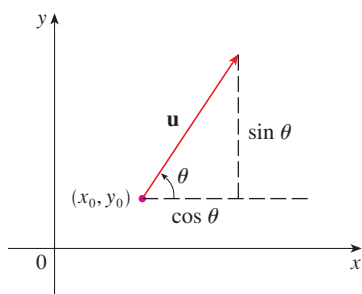


FIGURE 1

Directional Derivatives

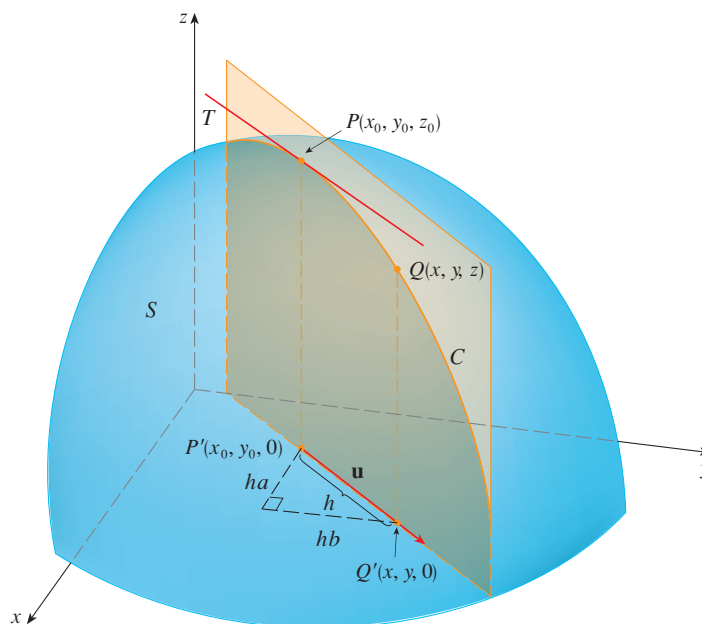
Recall that if $z = f(x, y)$, then the partial derivatives f_x and f_y are defined as

$$(1) \quad \begin{aligned} f_x(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h} \\ f_y(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h} \end{aligned}$$


FIGURE 2

A unit vector
 $\mathbf{u} = [a, b] = [\cos \theta, \sin \theta]$

TEC Visual 9.5A animates Figure 3 by rotating $\mathbf{u}$ and therefore T .


FIGURE 3

If $Q(x, y, z)$ is another point on C and P', Q' are the projections of P, Q onto the xy -plane, then the vector $\overrightarrow{P'Q'}$ is parallel to $\mathbf{u}$ and so

$$\overrightarrow{P'Q'} = h\mathbf{u} = [ha, hb]$$

for some scalar h . Therefore $x - x_0 = ha$, $y - y_0 = hb$, so $x = x_0 + ha$, $y = y_0 + hb$, and

$$\frac{\Delta z}{h} = \frac{z - z_0}{h} = \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

If we take the limit as $h \rightarrow 0$, we obtain the rate of change of z (with respect to distance) in the direction of $\mathbf{u}$, which is called the directional derivative of f in the direction of $\mathbf{u}$.

(2) Definition The **directional derivative** of f at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = [a, b]$ is

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

if this limit exists.

By comparing Definition 2 with Equations 1 we see that if $\mathbf{u} = \mathbf{i} = [1, 0]$, then $D_{\mathbf{i}}f = f_x$ and if $\mathbf{u} = \mathbf{j} = [0, 1]$, then $D_{\mathbf{j}}f = f_y$. In other words, the partial derivatives of f with respect to x and y are just special cases of the directional derivative.

EXAMPLE 1 | Use the weather map in Figure 1 to estimate the value of the directional derivative of the temperature function at Reno in the southeasterly direction.

SOLUTION The unit vector directed toward the southeast is $\mathbf{u} = [1/\sqrt{2}, -1/\sqrt{2}]$, but we won't need to use this expression. We start by drawing a line through Reno toward the southeast (see Figure 4).



FIGURE 4

We approximate the directional derivative $D_{\mathbf{u}}T$ by the average rate of change of the temperature between the points where this line intersects the isothermals $T = 50$ and $T = 60$. The temperature at the point southeast of Reno is $T = 60^\circ\text{F}$ and the temperature at the point northwest of Reno is $T = 50^\circ\text{F}$. The distance between these points looks to be about 75 miles. So the rate of change of the temperature in the southeasterly direction is

$$D_{\mathbf{u}}T \approx \frac{60 - 50}{75} = \frac{10}{75} \approx 0.13^\circ\text{F/mi}$$

When we compute the directional derivative of a function defined by a formula, we generally use the following theorem.

(3) Theorem If f is a differentiable function of x and y , then f has a directional derivative in the direction of any unit vector $\mathbf{u} = [a, b]$ and

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b$$

PROOF If we define a function g of the single variable h by

$$g(h) = f(x_0 + ha, y_0 + hb)$$

then, by the definition of a derivative, we have

$$\begin{aligned} (4) \quad g'(0) &= \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h} \\ &= D_{\mathbf{u}}f(x_0, y_0) \end{aligned}$$

On the other hand, we can write $g(h) = f(x, y)$, where $x = x_0 + ha$, $y = y_0 + hb$, so the Chain Rule (Theorem 9.4.2) gives

$$g'(h) = \frac{\partial f}{\partial x} \frac{dx}{dh} + \frac{\partial f}{\partial y} \frac{dy}{dh} = f_x(x, y)a + f_y(x, y)b$$

If we now put $h = 0$, then $x = x_0$, $y = y_0$, and

$$(5) \quad g'(0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b$$

Comparing Equations 4 and 5, we see that

$$D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b$$

If the unit vector $\mathbf{u}$ makes an angle θ with the positive x -axis (as in Figure 2), then we can write $\mathbf{u} = [\cos \theta, \sin \theta]$ and the formula in Theorem 3 becomes

$$(6) \quad D_{\mathbf{u}}f(x, y) = f_x(x, y) \cos \theta + f_y(x, y) \sin \theta$$

The directional derivative $D_{\mathbf{u}}f(1, 2)$ in Example 2 represents the rate of change of z in the direction of $\mathbf{u}$. This is the slope of the tangent line to the curve of intersection of the surface $z = x^3 - 3xy + 4y^2$ and the vertical plane through $(1, 2, 0)$ in the direction of $\mathbf{u}$ shown in Figure 5.

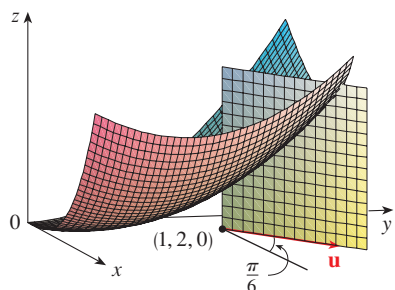


FIGURE 5

EXAMPLE 2 | Find the directional derivative $D_{\mathbf{u}}f(x, y)$ if

$$f(x, y) = x^3 - 3xy + 4y^2$$

and $\mathbf{u}$ is the unit vector given by angle $\theta = \pi/6$. What is $D_{\mathbf{u}}f(1, 2)$?

SOLUTION Formula 6 gives

$$\begin{aligned} D_{\mathbf{u}}f(x, y) &= f_x(x, y) \cos \frac{\pi}{6} + f_y(x, y) \sin \frac{\pi}{6} \\ &= (3x^2 - 3y) \frac{\sqrt{3}}{2} + (-3x + 8y) \frac{1}{2} \\ &= \frac{1}{2} [3\sqrt{3}x^2 - 3x + (8 - 3\sqrt{3})y] \end{aligned}$$

Therefore

$$D_{\mathbf{u}}f(1, 2) = \frac{1}{2} [3\sqrt{3}(1)^2 - 3(1) + (8 - 3\sqrt{3})(2)] = \frac{13 - 3\sqrt{3}}{2}$$

■ The Gradient Vector

Notice from Theorem 3 that the directional derivative of a differentiable function can be written as the dot product of two vectors:

$$\begin{aligned} (7) \quad D_{\mathbf{u}}f(x, y) &= f_x(x, y)a + f_y(x, y)b \\ &= [f_x(x, y), f_y(x, y)] \cdot [a, b] \\ &= [f_x(x, y), f_y(x, y)] \cdot \mathbf{u} \end{aligned}$$

The first vector in this dot product occurs not only in computing directional derivatives but in many other contexts as well. So we give it a special name (the *gradient* of f) and a special notation (**grad** f or ∇f , which is read “del f ”).

(8) Definition If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) = [f_x(x, y), f_y(x, y)]$$

EXAMPLE 3 | If $f(x, y) = \sin x + e^{xy}$, then

$$\nabla f(x, y) = [f_x, f_y] = [\cos x + ye^{xy}, xe^{xy}]$$

and

$$\nabla f(0, 1) = [2, 0]$$

With this notation for the gradient vector, we can rewrite the expression (7) for the directional derivative of a differentiable function as

(9)

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}$$

This expresses the directional derivative in the direction of $\mathbf{u}$ as the scalar projection of the gradient vector onto $\mathbf{u}$.

The gradient vector $\nabla f(2, -1)$ in Example 4 is shown in Figure 6 with initial point $(2, -1)$. Also shown is the vector $\mathbf{v}$ that gives the direction of the directional derivative. Both of these vectors are superimposed on a contour plot of the graph of f .

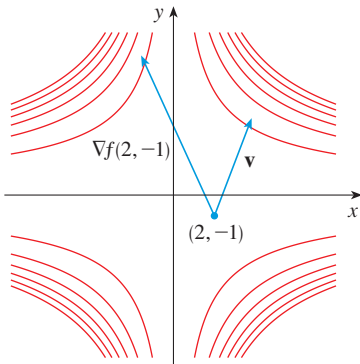


FIGURE 6

EXAMPLE 4 | Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction of the vector $\mathbf{v} = [2, 5]$.

SOLUTION We first compute the gradient vector at $(2, -1)$:

$$\nabla f(x, y) = [2xy^3, 3x^2y^2 - 4]$$

$$\nabla f(2, -1) = [-4, 8]$$

Note that $\mathbf{v}$ is not a unit vector, but since $|\mathbf{v}| = \sqrt{29}$, the unit vector in the direction of $\mathbf{v}$ is

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left[\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right]$$

Therefore, by Equation 9, we have

$$\begin{aligned} D_{\mathbf{u}}f(2, -1) &= \nabla f(2, -1) \cdot \mathbf{u} = [-4, 8] \cdot \left[\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right] \\ &= \frac{-4 \cdot 2 + 8 \cdot 5}{\sqrt{29}} = \frac{32}{\sqrt{29}} \end{aligned}$$

EXAMPLE 5 | **BB** Snake reversals and stripes In a study of the survivorship of juvenile garter snakes, a researcher¹ arrived at the model

$$F = 4.2 + 0.008R + 0.102S + 0.017R^2 - 0.034S^2 - 0.268RS$$

where F is a measure of the fitness of the snake, R measures the number of reversals of direction during flight from a predator, and S measures the degree of stripedness in the

1. E. Brodie III, "Correlational Selection for Color Pattern and Antipredator Behavior in the Garter Snake *Thamnophis ordinoides*," *Evolution* 46 (1992): 1284–98.

color pattern of the snake. We have previously considered how F changes as R or S changes by computing the partial derivatives with respect to R and S (Exercise 9.2.48). How does F change when $R = 3$ and $S = 2$ if the phenotype changes so that R and S increase by equal amounts?

SOLUTION The gradient vector of F is

$$\nabla F(R, S) = [F_R, F_S] = [0.008 + 0.034R - 0.268S, 0.102 - 0.068S - 0.268R]$$

and when $R = 3$ and $S = 2$ this becomes

$$\nabla F(3, 2) = [-0.426, -0.838]$$

We want the derivative in the direction halfway between $\mathbf{i}$ and $\mathbf{j}$, that is, in the direction $\mathbf{v} = [1, 1]$. The unit vector in this direction is $\mathbf{u} = (1/\sqrt{2})[1, 1]$, so the directional derivative is

$$\begin{aligned} D_{\mathbf{u}}F(3, 2) &= \nabla F(3, 2) \cdot \mathbf{u} = [-0.426, -0.838] \cdot \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \\ &= -\frac{1.264}{\sqrt{2}} \approx -0.894 \end{aligned}$$

This means that the fitness function at $(3, 2)$ decreases at a rate of 0.894 units when R and S increase by equal amounts. ■

■ Maximizing the Directional Derivative

Suppose we have a function f of two or three variables and we consider all possible directional derivatives of f at a given point. These give the rates of change of f in all possible directions. We can then ask the questions: In which of these directions does f change fastest and what is the maximum rate of change? The answers are provided by the following theorem.

TEC Visual 9.5B provides visual confirmation of Theorem 10.

(10) Theorem If f is a differentiable function and (a, b) is in the domain of f , then the maximum value of the directional derivative $D_{\mathbf{u}}f(a, b)$ is $|\nabla f(a, b)|$ and it occurs when $\mathbf{u}$ has the same direction as the gradient vector $\nabla f(a, b)$.

PROOF From Equation 9 we have

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| |\mathbf{u}| \cos \theta = |\nabla f| \cos \theta$$

where θ is the angle between ∇f and $\mathbf{u}$. The maximum value of $\cos \theta$ is 1 and this occurs when $\theta = 0$. Therefore the maximum value of $D_{\mathbf{u}}f$ is $|\nabla f|$ and it occurs when $\theta = 0$, that is, when $\mathbf{u}$ has the same direction as ∇f . ■

EXAMPLE 6

- If $f(x, y) = xe^y$, find the rate of change of f at the point $P(2, 0)$ in the direction from P to $Q(\frac{1}{2}, 2)$.
- In what direction does f have the maximum rate of change? What is this maximum rate of change?

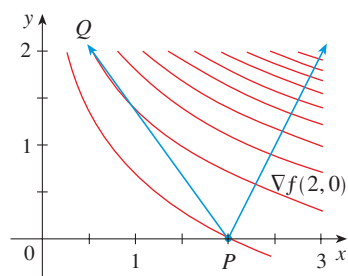


FIGURE 7

At $(2, 0)$ the function in Example 6 increases fastest in the direction of the gradient vector $\nabla f(2, 0) = [1, 2]$. Notice from Figure 7 that this vector appears to be perpendicular to the level curve through $(2, 0)$. Figure 8 shows the graph of f and the gradient vector.

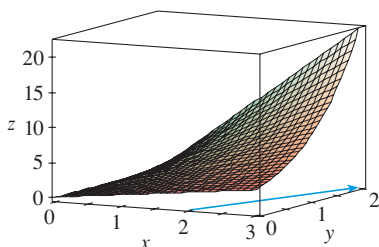


FIGURE 8

SOLUTION

(a) We first compute the gradient vector:

$$\nabla f(x, y) = [f_x, f_y] = [e^y, xe^y]$$

$$\nabla f(2, 0) = [1, 2]$$

The unit vector in the direction of $\overrightarrow{PQ} = [-1.5, 2]$ is $\mathbf{u} = \left[-\frac{3}{5}, \frac{4}{5}\right]$, so the rate of change of f in the direction from P to Q is

$$D_{\mathbf{u}}f(2, 0) = \nabla f(2, 0) \cdot \mathbf{u} = [1, 2] \cdot \left[-\frac{3}{5}, \frac{4}{5}\right]$$

$$= 1\left(-\frac{3}{5}\right) + 2\left(\frac{4}{5}\right) = 1$$

(b) According to Theorem 10, f increases fastest in the direction of the gradient vector $\nabla f(2, 0) = [1, 2]$. The maximum rate of change is

$$|\nabla f(2, 0)| = |[1, 2]| = \sqrt{5}$$

Another geometric aspect of the gradient vector is illustrated in Figure 9:

The gradient vector $\nabla f(a, b)$ is perpendicular to the level curve $f(x, y) = k$ that passes through the point (a, b) .

Although we won't prove this fact, it is intuitively plausible because the values of f remain constant as we move along the level curve and so it seems reasonable that the values of f should change most quickly in the perpendicular direction, which we know is the direction of $\nabla f(a, b)$ by Theorem 10.

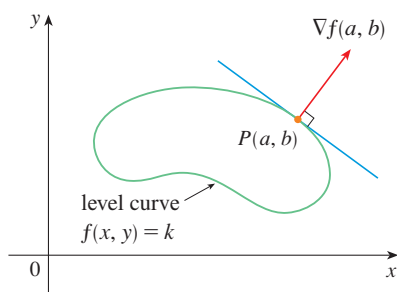


FIGURE 9

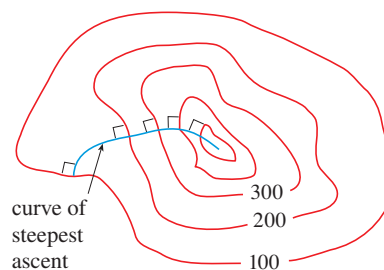
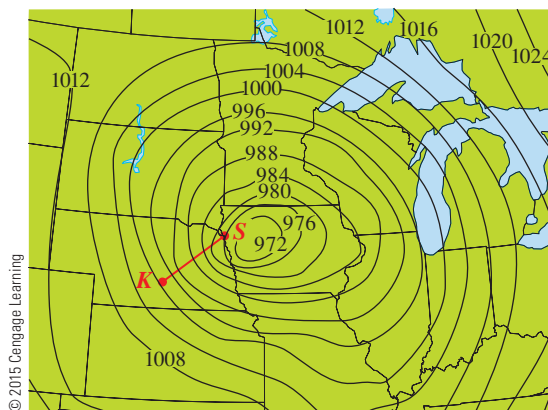


FIGURE 10

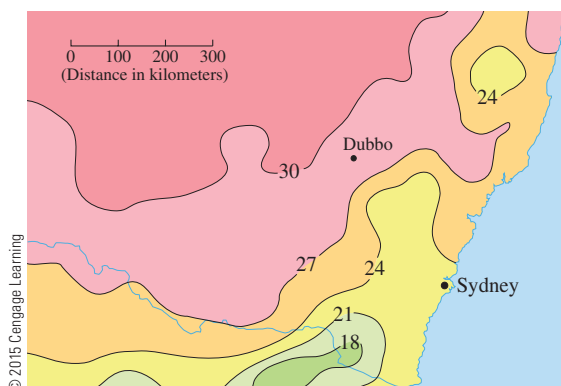
If we consider a topographical map of a hill and let $f(x, y)$ represent the height above sea level at a point with coordinates (x, y) , then a curve of steepest ascent can be drawn as in Figure 10 by making it perpendicular to all of the contour lines. This phenomenon can also be noticed in Figure 9.1.10, where Lonesome Creek follows a curve of steepest descent.

EXERCISES 9.5

1. Level curves for barometric pressure (in millibars) are shown for 6:00 AM on November 10, 1998. A deep low with pressure 972 mb is moving over northeast Iowa. The distance along the red line from K (Kearney, Nebraska) to S (Sioux City, Iowa) is 300 km. Estimate the value of the directional derivative of the pressure function at Kearney in the direction of Sioux City. What are the units of the directional derivative?



2. The contour map shows the average maximum temperature for November 2004 (in °C). Estimate the value of the directional derivative of this temperature function at Dubbo, New South Wales, in the direction of Sydney. What are the units?



3. A table of values for the wind-chill index $W = f(T, v)$ is given in Exercise 9.2.3. Use the table to estimate the value of $D_{\mathbf{u}} f(-20, 30)$, where $\mathbf{u} = [1/\sqrt{2}, 1/\sqrt{2}]$.

4–6 Find the directional derivative of f at the given point in the direction indicated by the angle θ .

4. $f(x, y) = x^2 y^3 - y^4$, $(2, 1)$, $\theta = \pi/4$
 5. $f(x, y) = ye^{-x}$, $(0, 4)$, $\theta = 2\pi/3$

6. $f(x, y) = x \sin(xy)$, $(2, 0)$, $\theta = \pi/3$

7–10

- (a) Find the gradient of f .
 (b) Evaluate the gradient at the point P .
 (c) Find the rate of change of f at P in the direction of the vector $\mathbf{u}$.

7. $f(x, y) = 5xy^2 - 4x^3y$, $P(1, 2)$, $\mathbf{u} = [\frac{5}{13}, \frac{12}{13}]$

8. $f(x, y) = y \ln x$, $P(1, -3)$, $\mathbf{u} = [-\frac{4}{5}, \frac{3}{5}]$

9. $f(x, y) = \sin(2x + 3y)$, $P(-6, 4)$, $\mathbf{u} = [\frac{1}{2}\sqrt{3}, -\frac{1}{2}]$

10. $f(x, y) = y^2/x$, $P(1, 2)$, $\mathbf{u} = [\frac{2}{3}, \frac{1}{3}\sqrt{5}]$

11–15 Find the directional derivative of the function at the given point in the direction of the vector $\mathbf{v}$.

11. $f(x, y) = 1 + 2x\sqrt{y}$, $(3, 4)$, $\mathbf{v} = [4, -3]$

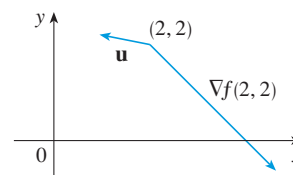
12. $f(x, y) = \ln(x^2 + y^2)$, $(2, 1)$, $\mathbf{v} = [-1, 2]$

13. $g(p, q) = p^4 - p^2q^3$, $(2, 1)$, $\mathbf{v} = [1, 3]$

14. $g(r, s) = \tan^{-1}(rs)$, $(1, 2)$, $\mathbf{v} = [5, 10]$

15. $V(u, t) = e^{-ut}$, $(0, 3)$, $\mathbf{v} = [2, -1]$

16. Use the figure to estimate $D_{\mathbf{u}} f(2, 2)$.



17. Find the directional derivative of $f(x, y) = \sqrt{xy}$ at $P(2, 8)$ in the direction of $Q(5, 4)$.

18. Find the directional derivative of $g(r, \theta) = e^{-r} \sin \theta$ at $P(0, \pi/3)$ in the direction of $Q(1, \pi/2)$.

19–22 Find the maximum rate of change of f at the given point and the direction in which it occurs.

19. $f(x, y) = y^2/x$, $(2, 4)$

20. $g(x, y) = x^2 - xy + y^2$, $(2, 1)$

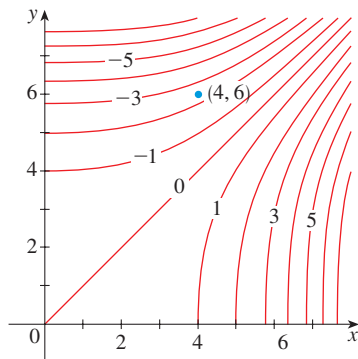
21. $f(x, y) = \sin(xy)$, $(1, 0)$

22. $f(p, q) = qe^{-p} + pe^{-q}$, $(0, 0)$

23. (a) Show that a differentiable function f decreases most rapidly at (a, b) in the direction opposite to the gradient vector, that is, in the direction of $-\nabla f(a, b)$.

- (b) Use the result of part (a) to find the direction in which the function $f(x, y) = x^4y - x^2y^3$ decreases fastest at the point $(2, -3)$.

- 24. Snake reversals and stripes** In Example 5 we calculated $\nabla F(3, 2)$, the gradient vector of the snake fitness function F when $R = 3$ and $S = 2$. In which direction $\mathbf{u}$ is the directional derivative of F at $(3, 2)$ a maximum? Calculate and interpret the maximum value.
- 25.** Find all points at which the direction of fastest change of the function $f(x, y) = x^2 + y^2 - 2x - 4y$ is $[1, 1]$.
- 26.** Sketch the gradient vector $\nabla f(4, 6)$ for the function f whose level curves are shown. Explain how you chose the direction and length of this vector.

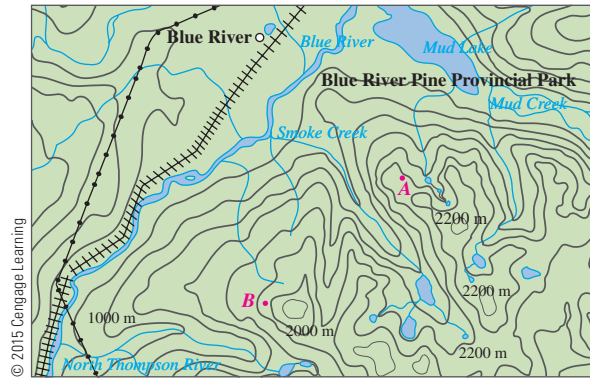


- 27.** Near a buoy, the depth of a lake at the point with coordinates (x, y) is $z = 200 + 0.02x^2 - 0.001y^3$, where x , y , and z are measured in meters. A fisherman in a small boat starts at the point $(80, 60)$ and moves toward the buoy, which is located at $(0, 0)$. Is the water under the boat getting deeper or shallower when he departs? Explain.
- 28.** Suppose you are climbing a hill whose shape is given by the equation $z = 1000 - 0.005x^2 - 0.01y^2$, where x , y , and z are measured in meters, and you are standing at a point with coordinates $(60, 40, 966)$. The positive x -axis points east and the positive y -axis points north.
- If you walk due south, will you start to ascend or descend? At what rate?
 - If you walk northwest, will you start to ascend or descend? At what rate?
 - In which direction is the slope largest? What is the rate of ascent in that direction? At what angle above the horizontal does the path in that direction begin?
- 29. Chemotaxis** is the phenomenon in which organisms move toward or away from certain chemicals. Suppose that a bacterium finds food by always moving toward the highest concentration of glucose. If it starts at the point $(1, 2)$ in the xy -plane and the concentration of glucose, in certain units, is

$$f(x, y) = \frac{1}{1 + x^2 + y^2}$$

find the direction in which the bacterium will move initially.

- 30.** Shown is a topographic map of Blue River Pine Provincial Park in British Columbia. Draw curves of steepest descent from point A (descending to Mud Lake) and from point B.



- 31. Controlling SARS** Example 9.1.12 used the equation

$$R_0(d, v) = 5(1 - v) \frac{d}{1 + d}$$

to model the effect of quarantine and vaccination on the spread of SARS, where v is the fraction of the population that is vaccinated and d is the average number of days an infectious individual remains in the population (recall that R_0 is the average number of new infections generated by each infected individual). Suppose that $d = 2$, $v = 0.1$, and public health authorities are considering a reallocation of resources that will increase v (which decreases R_0) but that comes at the cost of also increasing d (which increases R_0). The net effect of this reallocation on R_0 will therefore depend on how much the reallocation increases each of v and d . Suppose any reallocation results in d increasing twice as much as v .

- Use directional derivatives to determine whether a small reallocation of resources will be beneficial (that is, whether it will decrease R_0).
- Illustrate your answer to part (a) with a sketch of the directional derivative on a plot of the level curves of R_0 .

- 32. Growth and blood flow** The resistance of blood flowing through an artery is

$$R = C \frac{L}{r^4}$$

where L and r are the length and radius of the artery and C is a positive constant. Both L and r increase during growth. Suppose $r = 0.1$ mm, $L = 1$ mm, and $C = 1$.

- Suppose the length increases 10 mm for every mm increase in radius during growth. Use a directional derivative to determine the rate at which the resistance of blood flow changes with respect to a unit of growth in the rL -plane.
- Use a directional derivative to determine how much faster the length of the artery can change relative to that

- of its radius before the rate of change of resistance with respect to growth will be positive.
- (c) Illustrate your answers to parts (a) and (b) with a sketch of the directional derivatives on a plot of the level curves of R .
33. If $f(x, y) = xy$, find the gradient vector $\nabla f(3, 2)$ and use it to find the tangent line to the level curve $f(x, y) = 6$ at the point $(3, 2)$. Sketch the level curve, the tangent line, and the gradient vector.
34. If $g(x, y) = x^2 + y^2 - 4x$, find the gradient vector $\nabla g(1, 2)$ and use it to find the tangent line to the level curve $g(x, y) = 1$ at the point $(1, 2)$. Sketch the level curve, the tangent line, and the gradient vector.

35. The second directional derivative of $f(x, y)$ is

$$D_{\mathbf{u}}^2 f(x, y) = D_{\mathbf{u}}[D_{\mathbf{u}} f(x, y)]$$

If $f(x, y) = x^3 + 5x^2y + y^3$ and $\mathbf{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$, calculate $D_{\mathbf{u}}^2 f(2, 1)$.

36. (a) If $\mathbf{u} = [a, b]$ is a unit vector and f has continuous second partial derivatives, show that

$$D_{\mathbf{u}}^2 f = f_{xx}a^2 + 2f_{xy}ab + f_{yy}b^2$$

- (b) Find the second directional derivative of $f(x, y) = xe^{2y}$ in the direction of $\mathbf{v} = [4, 6]$.

9.6 Maximum and Minimum Values

As we saw in Chapter 4, one of the main uses of ordinary derivatives is in finding maximum and minimum values (extreme values). In this section we see how to use partial derivatives to locate maxima and minima of functions of two variables. In particular, in Example 6 we use Brodie's model to determine whether the survivorship function for snakes has a maximum value.

Look at the hills and valleys in the graph of f shown in Figure 1. There are two points (a, b) where f has a *local maximum*, that is, where $f(a, b)$ is larger than nearby values of $f(x, y)$. The larger of these two values is the *absolute maximum*. Likewise, f has two *local minima*, where $f(a, b)$ is smaller than nearby values. The smaller of these two values is the *absolute minimum*.

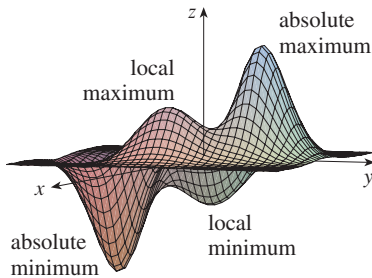


FIGURE 1

(1) Definition A function of two variables has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b) . [This means that $f(x, y) \leq f(a, b)$ for all points (x, y) in some disk with center (a, b) .] The number $f(a, b)$ is called a **local maximum value**. If $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b) , then f has a **local minimum** at (a, b) and $f(a, b)$ is a **local minimum value**.

If the inequalities in Definition 1 hold for all points (x, y) in the domain of f , then f has an **absolute maximum** (or **absolute minimum**) at (a, b) . An absolute maximum or minimum is sometimes called a **global** maximum or minimum.

(2) Fermat's Theorem for Functions of Two Variables If f has a local maximum or minimum at (a, b) and the first-order partial derivatives of f exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Notice that the conclusion of Theorem 2 can be stated in the notation of gradient vectors as $\nabla f(a, b) = \mathbf{0}$.

PROOF Let $g(x) = f(x, b)$. If f has a local maximum (or minimum) at (a, b) , then g has a local maximum (or minimum) at a , so $g'(a) = 0$ by Fermat's Theorem for functions of one variable (see Theorem 4.1.4). But $g'(a) = f_x(a, b)$ (see Equation 9.2.1) and so $f_x(a, b) = 0$. Similarly, by applying Fermat's Theorem to the function $G(y) = f(a, y)$, we obtain $f_y(a, b) = 0$. ■

If we put $f_x(a, b) = 0$ and $f_y(a, b) = 0$ in the equation of a tangent plane (Equation 9.3.2), we get $z = z_0$. Thus the geometric interpretation of Theorem 2 is that if the graph of f has a tangent plane at a local maximum or minimum, then the tangent plane must be horizontal.

A point (a, b) is called a **critical point** (or *stationary point*) of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if one of these partial derivatives does not exist. Theorem 2 says that if f has a local maximum or minimum at (a, b) , then (a, b) is a critical point of f . However, as in single-variable calculus, not all critical points give rise to maxima or minima. At a critical point, a function could have a local maximum or a local minimum or neither.

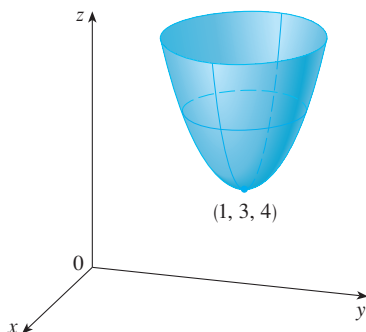


FIGURE 2
 $z = x^2 + y^2 - 2x - 6y + 14$

EXAMPLE 1 | Let $f(x, y) = x^2 + y^2 - 2x - 6y + 14$. Then

$$f_x(x, y) = 2x - 2 \quad f_y(x, y) = 2y - 6$$

These partial derivatives are equal to 0 when $x = 1$ and $y = 3$, so the only critical point is $(1, 3)$. By completing the square, we find that

$$f(x, y) = 4 + (x - 1)^2 + (y - 3)^2$$

Since $(x - 1)^2 \geq 0$ and $(y - 3)^2 \geq 0$, we have $f(x, y) \geq 4$ for all values of x and y . Therefore $f(1, 3) = 4$ is a local minimum, and in fact it is the absolute minimum of f . This can be confirmed geometrically from the graph of f , which is the paraboloid with vertex $(1, 3, 4)$ shown in Figure 2. ■

EXAMPLE 2 | Find the extreme values of $f(x, y) = y^2 - x^2$.

SOLUTION Since $f_x = -2x$ and $f_y = 2y$, the only critical point is $(0, 0)$. Notice that for points on the x -axis we have $y = 0$, so $f(x, y) = -x^2 < 0$ (if $x \neq 0$). However, for points on the y -axis we have $x = 0$, so $f(x, y) = y^2 > 0$ (if $y \neq 0$). Thus every disk with center $(0, 0)$ contains points where f takes positive values as well as points where f takes negative values. Therefore $f(0, 0) = 0$ can't be an extreme value for f , so f has no extreme value. ■

Example 2 illustrates the fact that a function need not have a maximum or minimum value at a critical point. Figure 3 shows how this is possible. The graph of f is the hyperbolic paraboloid $z = y^2 - x^2$, which has a horizontal tangent plane ($z = 0$) at the origin. You can see that $f(0, 0) = 0$ is a maximum in the direction of the x -axis but a minimum in the direction of the y -axis. Near the origin the graph has the shape of a saddle and so $(0, 0)$ is called a *saddle point* of f .

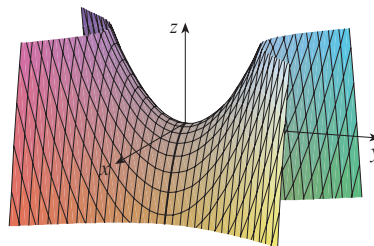


FIGURE 3
 $z = y^2 - x^2$

We need to be able to determine whether or not a function has an extreme value at a critical point. The following test, which is proved in Appendix E, is analogous to the Second Derivative Test for functions of one variable.

(3) Second Derivatives Test Suppose the second partial derivatives of f are continuous on a disk with center (a, b) , and suppose that $f_x(a, b) = 0$ and $f_y(a, b) = 0$ [that is, (a, b) is a critical point of f]. Let

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

- (a) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
- (b) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
- (c) If $D < 0$, then $f(a, b)$ is not a local maximum or minimum.

NOTE 1 In case (c) the point (a, b) is called a **saddle point** of f and the graph of f crosses its tangent plane at (a, b) .

NOTE 2 If $D = 0$, the test gives no information: f could have a local maximum or local minimum at (a, b) , or (a, b) could be a saddle point of f .

NOTE 3 To remember the formula for D , it's helpful to write it as a determinant:

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$$

EXAMPLE 3 | Find the local maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.

SOLUTION We first locate the critical points:

$$f_x = 4x^3 - 4y \quad f_y = 4y^3 - 4x$$

Setting these partial derivatives equal to 0, we obtain the equations

$$x^3 - y = 0 \quad \text{and} \quad y^3 - x = 0$$

To solve these equations we substitute $y = x^3$ from the first equation into the second one. This gives

$$0 = x^9 - x = x(x^8 - 1) = x(x^4 - 1)(x^4 + 1) = x(x^2 - 1)(x^2 + 1)(x^4 + 1)$$

so there are three real roots: $x = 0, 1, -1$. The three critical points are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

Next we calculate the second partial derivatives and $D(x, y)$:

$$f_{xx} = 12x^2 \quad f_{xy} = -4 \quad f_{yy} = 12y^2$$

$$D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = 144x^2y^2 - 16$$

Since $D(0, 0) = -16 < 0$, it follows from case (c) of the Second Derivatives Test that the origin is a saddle point; that is, f has no local maximum or minimum at $(0, 0)$. Since $D(1, 1) = 128 > 0$ and $f_{xx}(1, 1) = 12 > 0$, we see from case (a) of the test that $f(1, 1) = -1$ is a local minimum. Similarly, we have $D(-1, -1) = 128 > 0$ and $f_{xx}(-1, -1) = 12 > 0$, so $f(-1, -1) = -1$ is also a local minimum.

The graph of f is shown in Figure 4.

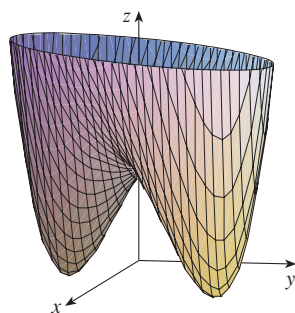


FIGURE 4
 $z = x^4 + y^4 - 4xy + 1$

A contour map of the function f in Example 3 is shown in Figure 5. The level curves near $(1, 1)$ and $(-1, -1)$ are oval in shape and indicate that as we move away from $(1, 1)$ or $(-1, -1)$ in any direction the values of f are increasing. The level curves near $(0, 0)$, on the other hand, resemble hyperbolas. They reveal that as we move away from the origin (where the value of f is 1), the values of f decrease in some directions but increase in other directions. Thus the contour map suggests the presence of the minima and saddle point that we found in Example 3.

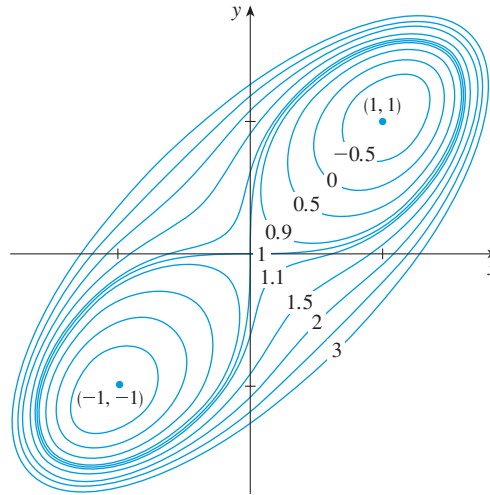


FIGURE 5

EXAMPLE 4 | Find the shortest distance from the point $(1, 0, -2)$ to the plane $x + 2y + z = 4$.

SOLUTION The distance from any point (x, y, z) to the point $(1, 0, -2)$ is

$$d = \sqrt{(x - 1)^2 + y^2 + (z + 2)^2}$$

but if (x, y, z) lies on the plane $x + 2y + z = 4$, then $z = 4 - x - 2y$ and so we have $d = \sqrt{(x - 1)^2 + y^2 + (6 - x - 2y)^2}$. We can minimize d by minimizing the simpler expression

$$d^2 = f(x, y) = (x - 1)^2 + y^2 + (6 - x - 2y)^2$$

By solving the equations

$$f_x = 2(x - 1) - 2(6 - x - 2y) = 4x + 4y - 14 = 0$$

$$f_y = 2y - 4(6 - x - 2y) = 4x + 10y - 24 = 0$$

we find that the only critical point is $(\frac{11}{6}, \frac{5}{3})$. Since $f_{xx} = 4$, $f_{xy} = 4$, and $f_{yy} = 10$, we have $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = 24 > 0$ and $f_{xx} > 0$, so by the Second Derivatives Test f has a local minimum at $(\frac{11}{6}, \frac{5}{3})$. Intuitively, we can see that this local minimum is actually an absolute minimum because there must be a point on the given plane that is closest to $(1, 0, -2)$. If $x = \frac{11}{6}$ and $y = \frac{5}{3}$, then

$$d = \sqrt{(x - 1)^2 + y^2 + (6 - x - 2y)^2} = \sqrt{(\frac{5}{6})^2 + (\frac{5}{3})^2 + (\frac{5}{6})^2} = \frac{5}{6}\sqrt{6}$$

The shortest distance from $(1, 0, -2)$ to the plane $x + 2y + z = 4$ is $\frac{5}{6}\sqrt{6}$. ■

EXAMPLE 5 | A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.

SOLUTION Let the length, width, and height of the box (in meters) be x , y , and z , as shown in Figure 6. Then the volume of the box is

$$V = xyz$$

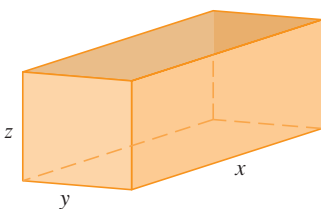


FIGURE 6

We can express V as a function of just two variables x and y by using the fact that the area of the four sides and the bottom of the box is

$$2xz + 2yz + xy = 12$$

Solving this equation for z , we get $z = (12 - xy)/[2(x + y)]$, so the expression for V becomes

$$V = xy \frac{12 - xy}{2(x + y)} = \frac{12xy - x^2y^2}{2(x + y)}$$

We compute the partial derivatives:

$$\frac{\partial V}{\partial x} = \frac{y^2(12 - 2xy - x^2)}{2(x + y)^2} \quad \frac{\partial V}{\partial y} = \frac{x^2(12 - 2xy - y^2)}{2(x + y)^2}$$

If V is a maximum, then $\partial V/\partial x = \partial V/\partial y = 0$, but $x = 0$ or $y = 0$ gives $V = 0$, so we must solve the equations

$$12 - 2xy - x^2 = 0 \quad 12 - 2xy - y^2 = 0$$

These imply that $x^2 = y^2$ and so $x = y$. (Note that x and y must both be positive in this problem.) If we put $x = y$ in either equation, we get $12 - 3x^2 = 0$, which gives $x = 2$, $y = 2$, and $z = (12 - 2 \cdot 2)/[2(2 + 2)] = 1$.

We could use the Second Derivatives Test to show that this gives a local maximum of V , or we could simply argue from the physical nature of this problem that there must be an absolute maximum volume, which has to occur at a critical point of V , so it must occur when $x = 2$, $y = 2$, $z = 1$. Then $V = 2 \cdot 2 \cdot 1 = 4$, so the maximum volume of the box is 4 m^3 . ■

EXAMPLE 6 | **BB** Snake reversals and stripes In Example 9.5.5 we investigated Brodie's model for the survivorship of young garter snakes:

$$F = 4.2 + 0.008R + 0.102S + 0.017R^2 - 0.034S^2 - 0.268RS$$

where F is a measure of the fitness of the snake, R measures the number of reversals of direction during flight from a predator, and S measures the degree of stripedness in the color pattern of the snake. Is there a maximum value of the fitness?

SOLUTION The partial derivatives of F are

$$F_R = 0.008 + 0.034R - 0.268S \quad F_S = 0.102 - 0.068S - 0.268R$$

To determine the critical point we need to solve the system of linear equations

$$0.034R - 0.268S = -0.008$$

$$0.268R + 0.068S = 0.102$$

Solving this system, we get $R \approx 0.361$, $S \approx 0.076$. The second-order derivatives are

$$F_{RR} = 0.034 \quad F_{RS} = -0.268 \quad F_{SS} = -0.068$$

So
$$D = F_{RR}F_{SS} - F_{RS}^2 = -0.074136$$

Because D is negative, the Second Derivatives Test tells us that the fitness function has no maximum or minimum value. This fact is illustrated by the graph and contour map of F in Figure 7. The function F has a saddle point at approximately $P(0.361, 0.076)$.

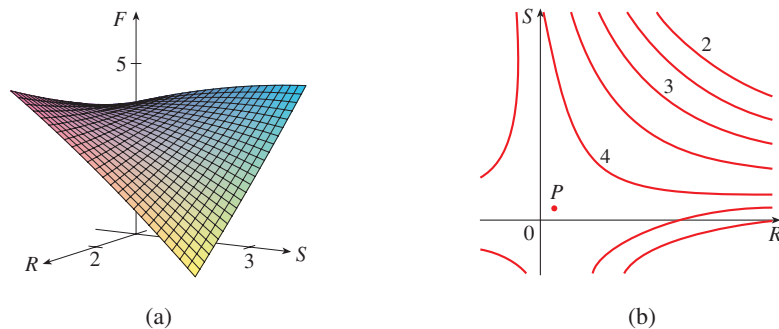


FIGURE 7

Fitness function for garter snakes

■ Absolute Maximum and Minimum Values

For a function f of one variable, the Extreme Value Theorem says that if f is continuous on a closed interval $[a, b]$, then f has an absolute minimum value and an absolute maximum value. According to the Closed Interval Method in Section 4.1, we found these by evaluating f not only at the critical numbers but also at the endpoints a and b .

There is a similar situation for functions of two variables. Just as a closed interval contains its endpoints, a **closed set** in $\mathbb{R}^2$ is one that contains all its boundary points. [A boundary point of D is a point (a, b) such that every disk with center (a, b) contains points in D and also points not in D .] For instance, the disk

$$D = \{(x, y) \mid x^2 + y^2 \leq 1\}$$

which consists of all points on and inside the circle $x^2 + y^2 = 1$, is a closed set because it contains all of its boundary points (which are the points on the circle $x^2 + y^2 = 1$). But if even one point on the boundary curve were omitted, the set would not be closed. (See Figure 8.)

A **bounded set** in $\mathbb{R}^2$ is one that is contained within some disk. In other words, it is finite in extent. Then, in terms of closed and bounded sets, we can state the following counterpart of the Extreme Value Theorem in two dimensions.

(4) Extreme Value Theorem for Functions of Two Variables If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D .

To find the extreme values guaranteed by Theorem 4, we note that, by Theorem 2, if f has an extreme value at (x_1, y_1) , then (x_1, y_1) is either a critical point of f or a boundary point of D . Thus we have the following extension of the Closed Interval Method.

(5) To find the absolute maximum and minimum values of a continuous function f on a closed, bounded set D :

1. Find the values of f at the critical points of f in D .
2. Find the extreme values of f on the boundary of D .
3. The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

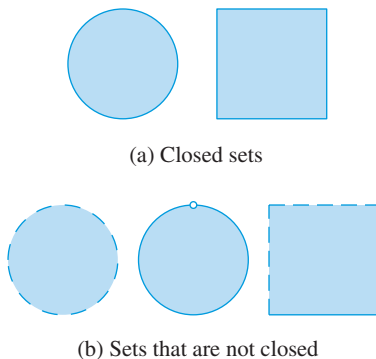


FIGURE 8

EXAMPLE 7 | Find the absolute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle $D = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$.

SOLUTION Since f is a polynomial, it is continuous on the closed, bounded rectangle D , so Theorem 4 tells us there is both an absolute maximum and an absolute minimum. According to step 1 in (5), we first find the critical points. These occur when

$$f_x = 2x - 2y = 0 \quad f_y = -2x + 2 = 0$$

so the only critical point is $(1, 1)$, and the value of f there is $f(1, 1) = 1$.

In step 2 we look at the values of f on the boundary of D , which consists of the four line segments L_1, L_2, L_3, L_4 shown in Figure 9. On L_1 we have $y = 0$ and

$$f(x, 0) = x^2 \quad 0 \leq x \leq 3$$

This is an increasing function of x , so its minimum value is $f(0, 0) = 0$ and its maximum value is $f(3, 0) = 9$. On L_2 we have $x = 3$ and

$$f(3, y) = 9 - 4y \quad 0 \leq y \leq 2$$

This is a decreasing function of y , so its maximum value is $f(3, 0) = 9$ and its minimum value is $f(3, 2) = 1$. On L_3 we have $y = 2$ and

$$f(x, 2) = x^2 - 4x + 4 \quad 0 \leq x \leq 3$$

By the methods of Chapter 4, or simply by observing that $f(x, 2) = (x - 2)^2$, we see that the minimum value of this function is $f(2, 2) = 0$ and the maximum value is $f(0, 2) = 4$. Finally, on L_4 we have $x = 0$ and

$$f(0, y) = 2y \quad 0 \leq y \leq 2$$

with maximum value $f(0, 2) = 4$ and minimum value $f(0, 0) = 0$. Thus, on the boundary, the minimum value of f is 0 and the maximum is 9.

In step 3 we compare these values with the value $f(1, 1) = 1$ at the critical point and conclude that the absolute maximum value of f on D is $f(3, 0) = 9$ and the absolute minimum value is $f(0, 0) = f(2, 2) = 0$. Figure 10 shows the graph of f .

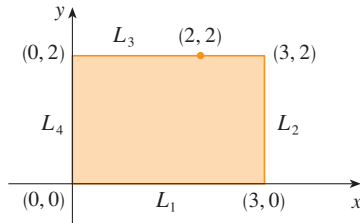


FIGURE 9

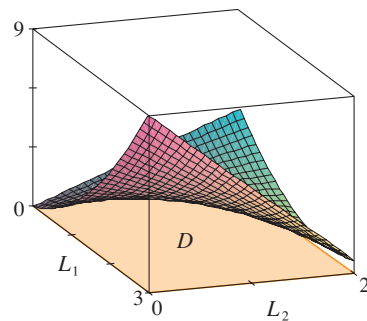


FIGURE 10
 $f(x, y) = x^2 - 2xy + 2y$

EXERCISES 9.6

1. Suppose $(1, 1)$ is a critical point of a function f with continuous second derivatives. In each case, what can you say about f ?

(a) $f_{xx}(1, 1) = 4$, $f_{xy}(1, 1) = 1$, $f_{yy}(1, 1) = 2$

(b) $f_{xx}(1, 1) = 4$, $f_{xy}(1, 1) = 3$, $f_{yy}(1, 1) = 2$

2. Suppose $(0, 2)$ is a critical point of a function g with continuous second derivatives. In each case, what can you say about g ?

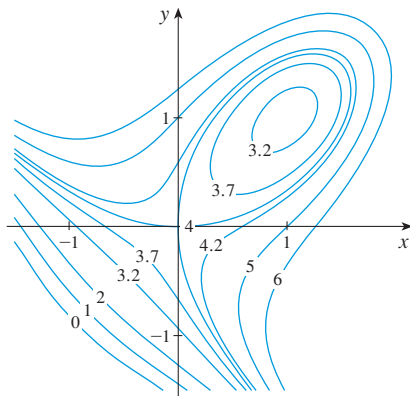
(a) $g_{xx}(0, 2) = -1$, $g_{xy}(0, 2) = 6$, $g_{yy}(0, 2) = 1$

(b) $g_{xx}(0, 2) = -1$, $g_{xy}(0, 2) = 2$, $g_{yy}(0, 2) = -8$

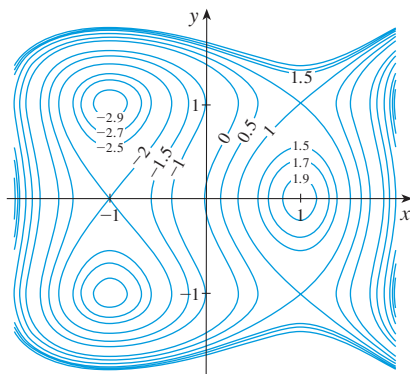
(c) $g_{xx}(0, 2) = 4$, $g_{xy}(0, 2) = 6$, $g_{yy}(0, 2) = 9$

3–4 Use the level curves in the figure to predict the location of the critical points of f and whether f has a saddle point or a local maximum or minimum at each critical point. Explain your reasoning. Then use the Second Derivatives Test to confirm your predictions.

3. $f(x, y) = 4 + x^3 + y^3 - 3xy$



4. $f(x, y) = 3x - x^3 - 2y^2 + y^4$



software, graph the function with a domain and viewpoint that reveal all the important aspects of the function.

5. $f(x, y) = x^2 + xy + y^2 + y$

6. $f(x, y) = x^3y + 12x^2 - 8y$

7. $f(x, y) = x^4 + y^4 - 4xy + 2$

8. $f(x, y) = xe^{-2x^2-2y^2}$

9. $f(x, y) = x^3 - 12xy + 8y^3$

10. $f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$

11. $f(x, y) = e^x \cos y$

12. $f(x, y) = y \cos x$

13. $f(x, y) = (x^2 + y^2)e^{y^2-x^2}$

14. $f(x, y) = e^y(y^2 - x^2)$

15. $f(x, y) = y^2 - 2y \cos x$, $-1 \leq x \leq 7$

16. $f(x, y) = \sin x \sin y$, $-\pi < x < \pi$, $-\pi < y < \pi$

17–20 Find the absolute maximum and minimum values of f on the set D .

17. $f(x, y) = 1 + 4x - 5y$, D is the closed triangular region with vertices $(0, 0)$, $(2, 0)$, and $(3, 0)$

18. $f(x, y) = 3 + xy - x - 2y$, D is the closed triangular region with vertices $(1, 0)$, $(5, 0)$, and $(1, 4)$

19. $f(x, y) = x^2 + y^2 + x^2y + 4$,
 $D = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}$

20. $f(x, y) = 4x + 6y - x^2 - y^2$,
 $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq 5\}$

-  **21.** For functions of one variable it is impossible for a continuous function to have two local maxima and no local minimum. But for functions of two variables such functions exist. Show that the function

$$f(x, y) = -(x^2 - 1)^2 - (x^2y - x - 1)^2$$

has only two critical points, but has local maxima at both of them. Then use a graphing device to produce a graph with a carefully chosen domain and viewpoint to see how this is possible.

-  **22.** If a function of one variable is continuous on an interval and has only one critical number, then a local maximum has to be an absolute maximum. But this is not true for functions of two variables. Show that the function

$$f(x, y) = 3xe^y - x^3 - e^{3y}$$

has exactly one critical point, and that f has a local maximum

5–16 Find the local maximum and minimum values and saddle point(s) of the function. If you have three-dimensional graphing

there that is not an absolute maximum. Then use a graphing device to produce a graph with a carefully chosen domain and viewpoint to see how this is possible.

23. Find the shortest distance from the point $(2, 1, -1)$ to the plane $x + y - z = 1$.
24. Find the point on the plane $x - y + z = 4$ that is closest to the point $(1, 2, 3)$.
25. Find the points on the cone $z^2 = x^2 + y^2$ that are closest to the point $(4, 2, 0)$.
26. Find the points on the surface $y^2 = 9 + xz$ that are closest to the origin.
27. Find three positive numbers whose sum is 100 and whose product is a maximum.
28. Find three positive numbers whose sum is 12 and the sum of whose squares is as small as possible.
29. Find the maximum volume of a rectangular box that is inscribed in a sphere of radius r .
30. Find the dimensions of the box with volume 1000 cm^3 that has minimal surface area.
31. Find the volume of the largest rectangular box in the first octant with three faces in the coordinate planes and one vertex in the plane $x + 2y + 3z = 6$.
32. Find the dimensions of the rectangular box with largest volume if the total surface area is given as 64 cm^2 .
33. Find the dimensions of a rectangular box of maximum volume such that the sum of the lengths of its 12 edges is a constant c .
34. The base of an aquarium with given volume V is made of slate and the sides are made of glass. If slate costs five times as much (per unit area) as glass, find the dimensions of the aquarium that minimize the cost of the materials.
35. A cardboard box without a lid is to have a volume of $32,000 \text{ cm}^3$. Find the dimensions that minimize the amount of cardboard used.
36. **Crop yield** A model for the yield Y of an agricultural crop as a function of the nitrogen level N and phosphorus level P in the soil (measured in appropriate units) is

$$Y(N, P) = kNP e^{-N-P}$$

where k is a positive constant. What levels of nitrogen and phosphorus result in the best yield?

37. The **Shannon index** (sometimes called the Shannon-Wiener index or Shannon-Weaver index) is a measure of diversity in an ecosystem. For the case of three species, it is defined as

$$H = -p_1 \ln p_1 - p_2 \ln p_2 - p_3 \ln p_3$$

where p_i is the proportion of the ecosystem made up of species i .

- (a) Express H as a function of two variables using the fact that $p_1 + p_2 + p_3 = 1$.
 - (b) What is the domain of H ?
 - (c) Find the maximum value of H . For what values of p_1, p_2, p_3 does it occur?
38. A rectangular building is being designed to minimize heat loss. The east and west walls lose heat at a rate of 10 units/m^2 per day, the north and south walls at a rate of 8 units/m^2 per day, the floor at a rate of 1 unit/m^2 per day, and the roof at a rate of 5 units/m^2 per day. Each wall must be at least 30 m long, the height must be at least 4 m , and the volume must be exactly 4000 m^3 .
 - (a) Find and sketch the domain of the heat loss as a function of the lengths of the sides.
 - (b) Find the dimensions that minimize heat loss. (Check both the critical points and the points on the boundary of the domain.)
 - (c) Could you design a building with even less heat loss if the restrictions on the lengths of the walls were removed?
 39. **Infectious disease control** In Example 9.1.12 we considered a model for the spread of SARS:

$$R_0(d, v) = 5(1 - v) \frac{d}{1 + d}$$

where R_0 is the basic reproduction number, v is the fraction of the population that is vaccinated, and d is the average number of days that an infected individual remains in the population before being quarantined. If we restrict our attention to the case where $0 \leq d \leq 20$, what is the maximum value of R_0 ?

40. **Hardy-Weinberg Law** Three alleles (alternative versions of a gene) A, B, and O determine the four blood types A (AA or AO), B (BB or BO), O (OO), and AB. The Hardy-Weinberg Law states that the proportion of individuals in a population who carry two different alleles is

$$P = 2pq + 2pr + 2rq$$

where p, q , and r are the proportions of A, B, and O in the population. Use the fact that $p + q + r = 1$ to show that P is at most $\frac{2}{3}$.

Chapter 9 REVIEW

CONCEPT CHECK

1. (a) What is a function of two variables?
(b) Describe two methods for visualizing a function of two variables.
2. What does $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ mean? How can you show that such a limit does not exist?
3. (a) What does it mean to say that f is continuous at (a,b) ?
(b) If f is continuous on $\mathbb{R}^2$, what can you say about its graph?
4. (a) Write expressions for the partial derivatives $f_x(a,b)$ and $f_y(a,b)$ as limits.
(b) How do you interpret $f_x(a,b)$ and $f_y(a,b)$ geometrically? How do you interpret them as rates of change?
(c) If $f(x,y)$ is given by a formula, how do you calculate f_x and f_y ?
5. What does Clairaut's Theorem say?
6. How do you find an equation for the tangent plane to a surface $z = f(x,y)$?
7. Define the linearization of f at (a,b) . What is the corresponding linear approximation? What is the geometric interpretation of the linear approximation?
8. (a) What does it mean to say that f is differentiable at (a,b) ?
(b) How do you usually verify that f is differentiable?
9. State the Chain Rule for the case where $z = f(x,y)$ and x and y are functions of a variable t .
10. If z is defined implicitly as a function of x and y by an equation of the form $F(x,y,z) = 0$, how do you find $\partial z / \partial x$ and $\partial z / \partial y$?
11. (a) Write an expression as a limit for the directional derivative of f at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = [a, b]$. How do you interpret it as a rate? How do you interpret it geometrically?
(b) If f is differentiable, write an expression for $D_{\mathbf{u}}f(x_0, y_0)$ in terms of f_x and f_y .
12. (a) Define the gradient vector ∇f for a function f of two variables.
(b) Express $D_{\mathbf{u}}f$ in terms of ∇f .
(c) Explain the geometric significance of the gradient.
13. What do the following statements mean?
(a) f has a local maximum at (a,b) .
(b) f has an absolute maximum at (a,b) .
(c) f has a local minimum at (a,b) .
(d) f has an absolute minimum at (a,b) .
(e) f has a saddle point at (a,b) .
14. (a) If f has a local maximum at (a,b) , what can you say about its partial derivatives at (a,b) ?
(b) What is a critical point of f ?
15. State the Second Derivatives Test.
16. (a) What is a closed set in $\mathbb{R}^2$? What is a bounded set?
(b) State the Extreme Value Theorem for functions of two variables.
(c) How do you find the values that the Extreme Value Theorem guarantees?

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

1. $f_y(a,b) = \lim_{y \rightarrow b} \frac{f(a,y) - f(a,b)}{y - b}$
2. There exists a function f with continuous second-order partial derivatives such that $f_x(x,y) = x + y^2$ and $f_y(x,y) = x - y^2$.
3. $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$
4. $D_{\mathbf{u}}f(x,y,z) = f_z(x,y,z)$, where $\mathbf{u} = [0, 0, 1]$
5. If $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$ along every straight line through (a,b) , then $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$.
6. If $f_x(a,b)$ and $f_y(a,b)$ both exist, then f is differentiable at (a,b) .
7. If f has a local minimum at (a,b) and f is differentiable at (a,b) , then $\nabla f(a,b) = \mathbf{0}$.
8. If f is a function, then $\lim_{(x,y) \rightarrow (2,5)} f(x,y) = f(2,5)$
9. If $f(x,y) = \ln y$, then $\nabla f(x,y) = 1/y$.

10. If $(2, 1)$ is a critical point of f and

$$f_{xx}(2, 1)f_{yy}(2, 1) < [f_{xy}(2, 1)]^2$$

then f has a saddle point at $(2, 1)$.

11. If $f(x, y) = \sin x + \sin y$, then $-\sqrt{2} \leq D_{\mathbf{u}}f(x, y) \leq \sqrt{2}$.

12. If $f(x, y)$ has two local maxima, then f must have a local minimum.

EXERCISES

- 1–2** Find and sketch the domain of the function.

1. $f(x, y) = \ln(x + y + 1)$

2. $f(x, y) = \sqrt{4 - x^2 - y^2} + \sqrt{1 - x^2}$

- 3–4** Sketch the graph of the function.

3. $f(x, y) = 1 - y^2$

4. $f(x, y) = 1 - \frac{1}{5}x - \frac{1}{3}y$

- 5–6** Sketch several level curves of the function.

5. $f(x, y) = \ln(1 + x^2 + y^2)$ 6. $f(x, y) = e^x + y$

- 7–8** Evaluate the limit or show that it does not exist.

7. $\lim_{(x,y) \rightarrow (1,1)} \frac{2xy}{x^2 + 2y^2}$

8. $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2 + 2y^2}$

9. A metal plate is situated in the xy -plane and occupies the rectangle $0 \leq x \leq 10$, $0 \leq y \leq 8$, where x and y are measured in meters. The temperature at the point (x, y) in the plate is $T(x, y)$, where T is measured in degrees Celsius. Temperatures at equally spaced points were measured and recorded in the table.

- (a) Estimate the values of the partial derivatives $T_x(6, 4)$ and $T_y(6, 4)$. What are the units?
 (b) Estimate the value of $D_{\mathbf{u}}T(6, 4)$, where $\mathbf{u} = [1/\sqrt{2}, 1/\sqrt{2}]$. Interpret your result.
 (c) Estimate the value of $T_{xy}(6, 4)$.

$x \backslash y$	0	2	4	6	8
0	30	38	45	51	55
2	52	56	60	62	61
4	78	74	72	68	66
6	98	87	80	75	71
8	96	90	86	80	75
10	92	92	91	87	78

10. Find a linear approximation to the temperature function $T(x, y)$ in Exercise 9 near the point $(6, 4)$. Then use it to estimate the temperature at the point $(5, 3.8)$.

- 11–15** Find the first partial derivatives.

11. $f(x, y) = \sqrt{2x + y^2}$

12. $u = e^{-r} \sin 2\theta$

13. $g(u, v) = u \tan^{-1} v$

14. $w = \frac{x}{y - z}$

15. $T(p, q, r) = p \ln(q + e^r)$

- 16. Predators handling prey** C. S. Holling proposed the following model for the number of prey P_e eaten by a predator during a fixed time period:

$$P_e(N, T_h) = \frac{aN}{1 + aT_hN}$$

where a is a positive constant called the attack rate, N is the prey density, and T_h is the handling time. (Handling refers to catching the prey, moving it, eating it, and digesting it.) Calculate the partial derivatives $\partial P_e / \partial N$ and $\partial P_e / \partial T_h$ and interpret them. Are they positive or negative? What do you conclude? Are your conclusions reasonable?

Source: Adapted from C. Holling, "Some Characteristics of Simple Types of Predation and Parasitism," *The Canadian Entomologist* 91 (1959): 385–98.

- 17. Tadpole predation** In an experiment, the probability that a tadpole with mass m (in grams) will escape capture by predators is modeled as

$$P(m, T) = \frac{1}{1 + e^{3.222 - 31.669m + 0.083T}}$$

where T is the temperature of the water (in degrees Celsius).

- (a) Calculate $\partial P / \partial m$ and $\partial P / \partial T$ when $m = 0.2$ g and $T = 20^\circ\text{C}$.
 (b) Will increasing the water temperature improve or worsen the tadpoles' chance of escape?
 (c) As the tadpoles grow bigger, do they have a better chance of escape?

Source: Adapted from M. Anderson et al., "The Direct and Indirect Effects of Temperature on a Predator-Prey Relationship," *Canadian Journal of Zoology* 79 (2001): 1834–41.

18. The speed of sound traveling through ocean water is a function of temperature, salinity, and pressure. It has been modeled by the function

$$C = 1449.2 + 4.6T - 0.055T^2 + 0.00029T^3 \\ + (1.34 - 0.01T)(S - 35) + 0.016D$$

where C is the speed of sound (in meters per second), T is the temperature (in degrees Celsius), S is the salinity (the concentration of salts in parts per thousand, which means the number of grams of dissolved solids per 1000 g of water), and D is the depth below the ocean surface (in meters). Compute $\partial C/\partial T$, $\partial C/\partial S$, and $\partial C/\partial D$ when $T = 10^\circ\text{C}$, $S = 35$ parts per thousand, and $D = 100$ m. Explain the physical significance of these partial derivatives.

19–22 Find all second partial derivatives of f .

19. $f(x, y) = 4x^3 - xy^2$ 20. $z = xe^{-2y}$
 21. $f(x, y, z) = x^k y^l z^m$ 22. $v = r \cos(s + 2t)$

23. If $z = xy + xe^{y/x}$, show that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = xy + z$.

24. If $z = \sin(x + \sin t)$, show that

$$\frac{\partial z}{\partial x} \frac{\partial^2 z}{\partial x \partial t} = \frac{\partial z}{\partial t} \frac{\partial^2 z}{\partial x^2}$$

25–28 Find an equation of the tangent plane to the given surface at the specified point.

25. $z = 4x^2 - y^2 + 2y$, $(-1, 2, 4)$
 26. $z = 9x^2 + y^2 + 6x - 3y + 5$, $(1, 2, 18)$
 27. $z = \sqrt{4 - x^2 - 2y^2}$, $(1, -1, 1)$
 28. $z = y \ln x$, $(1, 4, 0)$

29. Explain why the function $f(x, y) = xy + y^3$ is differentiable. Then find the linearization of f at the point $(2, 1)$.

30. Find the linear approximation of the function $f(x, y, z) = x^3 \sqrt{y^2 + z^2}$ at the point $(2, 3, 4)$ and use it to estimate the number $(1.98)^3 \sqrt{(3.01)^2 + (3.97)^2}$.

31. If $u = x^2 y^3 + z^4$, where $x = p + 3p^2$, $y = pe^p$, and $z = p \sin p$, use the Chain Rule to find du/dp .

32. **Body surface area** A model for the surface area of a human body is given by the function

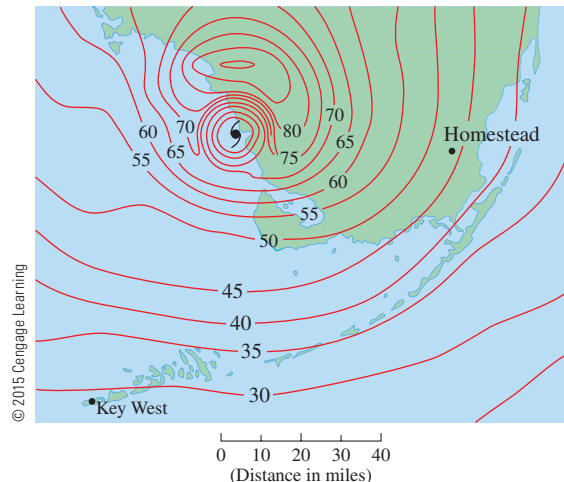
$$S = f(w, h) = 0.1091w^{0.425}h^{0.725}$$

where w is the weight (in pounds), h is the height (in inches), and S is measured in square feet. Both w and h depend on the age a of the person. Use the Chain Rule to find an expression for the rate of change of S with respect to a .

33. If $z = f(u, v)$, where $u = xy$, $v = y/x$, and f has continuous second partial derivatives, show that

$$x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = -4uv \frac{\partial^2 z}{\partial u \partial v} + 2v \frac{\partial z}{\partial v}$$

34. If $\cos(xyz) = 1 + x^2 y^2 + z^2$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 35. Find the gradient of the function $g(p, q) = pq^2 e^{-pq}$.
 36. (a) When is the directional derivative of f a maximum?
 (b) When is it a minimum?
 (c) When is it 0?
 (d) When is it half of its maximum value?
 37. Find the directional derivative of $f(x, y) = x^2 e^{-y}$ at the point $(-2, 0)$ in the direction toward the point $(2, -3)$.
 38. Find the maximum rate of change of $f(x, y) = x^2 y + \sqrt{y}$ at the point $(2, 1)$. In which direction does it occur?
 39. The contour map shows wind speed in knots during Hurricane Andrew on August 24, 1992. Use it to estimate the value of the directional derivative of the wind speed at Homestead, Florida, in the direction of the eye of the hurricane.



40–42 Find the local maximum and minimum values and saddle points of the function. If you have three-dimensional graphing software, graph the function with a domain and view-point that reveal all the important aspects of the function.

40. $f(x, y) = x^3 - 6xy + 8y^3$
 41. $f(x, y) = 3xy - x^2 y - xy^2$ 42. $f(x, y) = (x^2 + y)e^{y/2}$

43–44 Find the absolute maximum and minimum values of f on the set D .

43. $f(x, y) = 4xy^2 - x^2 y^2 - xy^3$; D is the closed triangular region in the xy -plane with vertices $(0, 0)$, $(0, 6)$, and $(6, 0)$
 44. $f(x, y) = e^{-x^2 - y^2}(x^2 + 2y^2)$; D is the disk $x^2 + y^2 \leq 4$

Systems of Linear Differential Equations

10

Jellyfish, like those shown here, move by contracting part of their body to create a high-pressure water jet. We model this phenomenon in Exercises 10.1.34 and 10.3.1.

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10.1 Qualitative Analysis of Linear Systems

10.2 Solving Systems of Linear Differential Equations

10.3 Applications

PROJECT: Pharmacokinetics of Antimicrobial Dosing

10.4 Systems of Nonlinear Differential Equations

CASE STUDY 2d: Hosts, Parasites, and Time-Travel

IN SECTIONS 7.5 AND 7.6 we introduced systems of differential equations. In this chapter we deal with a special kind of system called a system of linear differential equations. Such systems deserve special attention because they arise in a number of areas in the life sciences and because it is possible to develop a complete understanding of their behavior. As we will see in Section 10.4, they also form the foundation for analyzing systems of nonlinear differential equations.

10.1 Qualitative Analysis of Linear Systems

In this section we introduce systems of linear differential equations and give some examples of the kinds of behavior they can exhibit. Our focus is on two-dimensional systems.

Terminology

A two-dimensional **system of linear differential equations** has the form

$$\begin{aligned}\frac{dx_1}{dt} &= a_{11}(t)x_1 + a_{12}(t)x_2 + g_1(t) \\ \frac{dx_2}{dt} &= a_{21}(t)x_1 + a_{22}(t)x_2 + g_2(t)\end{aligned}$$

where $x_1(t)$ and $x_2(t)$ are unknown functions, $a_{ij}(t)$ are coefficients, and $g_i(t)$ are functions of time (sometimes called forcing, or input, functions). In matrix notation the system can be written

$$(1) \quad \frac{d\mathbf{x}}{dt} = A(t)\mathbf{x} + \mathbf{g}(t)$$

where $\mathbf{x}(t)$ is a column vector whose components are the unknown functions $x_1(t)$ and $x_2(t)$, $d\mathbf{x}/dt$ is a column vector whose components are dx_1/dt and dx_2/dt , $A(t)$ is a matrix whose entries are the coefficients, and $\mathbf{g}(t)$ is a column vector whose components are the forcing functions.

Using the terminology introduced in Section 7.1, we say that Equation 1 is a system of linear *first-order nonautonomous* differential equations. If A and $\mathbf{g}$ are independent of time, then Equation 1 is a system of linear first-order *autonomous* differential equations.

In this chapter we will study only autonomous differential equations. In fact, most of our considerations will also assume $\mathbf{g} = \mathbf{0}$. Such systems are called **homogeneous**. As shown in Exercise 23, any autonomous system of equations for which $\mathbf{g} \neq \mathbf{0}$ (which is called a **nonhomogeneous** system) can be reduced to a homogeneous system in which $\mathbf{g} = \mathbf{0}$ through a change of variables. Thus we consider systems of the form

$$\begin{aligned}\frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 \\ \frac{dx_2}{dt} &= a_{21}x_1 + a_{22}x_2\end{aligned}$$

or, in matrix notation,

$$(2) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

EXAMPLE 1 | Radioimmunotherapy is a cancer treatment in which radioactive atoms are attached to tumor-specific antibody molecules and then injected into the bloodstream. The antibody molecules then attach only to tumor cells, where they deliver the cell-killing radioactivity. Mathematical models have been used to optimize this treatment. Let's use $x_1(t)$ and $x_2(t)$ to denote the amount of antibody (in μg) in the blood and the tumor, respectively, at time t (in minutes after the start of treatment). If a denotes the per unit rate of clearance from the blood, b the per unit rate of movement from the blood into the tumor, and c the per unit rate of clearance from the tumor, a simple model is¹

$$\begin{aligned}\frac{dx_1}{dt} &= -ax_1 - bx_1 \\ \frac{dx_2}{dt} &= bx_1 - cx_2\end{aligned}$$

What is the matrix A for this model when it is written in the form of Equation 2?

SOLUTION Using $\mathbf{x}$ to denote the vector with components x_1 and x_2 , we can write this system as Equation 2 with

$$A = \begin{bmatrix} -(a+b) & 0 \\ b & -c \end{bmatrix}$$

EXAMPLE 2 | Metapopulation dynamics A population of deer mice is split into two patches through habitat fragmentation. The population in patch A reproduces at a per capita rate of 2, while that in patch B reproduces at a per capita rate of -1 (where time is measured in years). The reproductive rate in patch 2 is negative because the mortality rate in this patch is larger than the birth rate. The per capita movement rate from patch A to B is 3, and from patch B to A is 2. Write a system of two linear differential equations that describes the two patches.

SOLUTION Let's use $x_A(t)$ and $x_B(t)$ to denote the number of individuals in patches A and B, respectively. Individuals in patch A reproduce at a per capita rate of 2, and they leave the patch at a per capita rate of 3. At the same time, individuals migrate into the patch from patch B at a per capita rate of 2. Therefore,

$$\begin{aligned}\frac{dx_A}{dt} &= 2x_A - 3x_A + 2x_B \\ &= -x_A + 2x_B\end{aligned}$$

Similarly, individuals in patch B reproduce at a per capita rate of -1 , they leave the patch at per capita rate 2, and individuals from patch A enter the patch at per capita rate 3. We obtain

$$\begin{aligned}\frac{dx_B}{dt} &= -x_B - 2x_B + 3x_A \\ &= 3x_A - 3x_B\end{aligned}$$

Using $\mathbf{x}$ to denote the vector with components x_A and x_B , we can write this system in matrix notation in the form of Equation 2 with

$$A = \begin{bmatrix} -1 & 2 \\ 3 & -3 \end{bmatrix}$$

1. A. Flynn et al., "Effectiveness of Radiolabelled Antibodies for Radio-Immunotherapy in a Colorectal Xenograft Model: A Comparative Study using the Linear-Quadratic Formulation," *International Journal of Radiation Biology* 77 (2001): 507–17.

We can employ some of the techniques of phase-plane analysis introduced in Section 7.6 to systems of linear differential equations. This includes plotting the nullclines, finding equilibria, and assessing the stability of any equilibria.

We recall the definition of an equilibrium given in 7.6.2, but we make it specific to systems of two linear differential equations.

(3) Definition An **equilibrium** of the system of differential equations given by (2) is a pair of values $(\hat{x}_1, \hat{x}_2)$ such that both $dx_1/dt = 0$ and $dx_2/dt = 0$ when $x_1 = \hat{x}_1$ and $x_2 = \hat{x}_2$. Equivalently, from Equation 2, the vector of equilibrium values $\hat{\mathbf{x}}$ satisfies the equation $A\hat{\mathbf{x}} = \mathbf{0}$.

Theorem 8.6.5 tells us that the equation $A\hat{\mathbf{x}} = \mathbf{0}$ always has the solution $\hat{\mathbf{x}} = \mathbf{0}$. Thus the origin $(\hat{x}_1, \hat{x}_2) = (0, 0)$ is always an equilibrium of Equation 2. This theorem also says that, in the event that $\det A \neq 0$, the origin is the only equilibrium. This is referred to as the **generic** case. The case where $\det A = 0$ is referred to as **nongeneric**. In most of this chapter we will concentrate on the generic case because it is most common in life science applications.

Recall also the definition of a nullcline from Section 7.6. In the context of Equation 2, we have the following definition.

Definition The x_1 -nullcline of differential equation (2) is the set of points in the x_1x_2 -plane satisfying the equation $dx_1/dt = 0$. From Equation 2, this is the line defined by the equation $a_{11}x_1 + a_{12}x_2 = 0$. The x_2 -nullcline of differential equation (2) is the set of points in the x_1x_2 -plane satisfying the equation $dx_2/dt = 0$. From Equation 2, this is the line defined by the equation $a_{21}x_1 + a_{22}x_2 = 0$.

Throughout this chapter we use a single color (purple) for the nullclines of both variables.

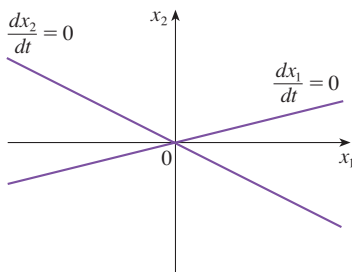


FIGURE 1

Thus a system of two linear differential equations has two nullclines, one for each variable. Both nullclines are straight lines and they intersect at the origin because both $dx_1/dt = 0$ and $dx_2/dt = 0$ at the point $(x_1, x_2) = (0, 0)$. (See Figure 1.) This is consistent with our earlier observation that the origin is an equilibrium.

EXAMPLE 3 | Metapopulation dynamics (continued) Determine the equilibrium of the model in Example 2 and interpret it biologically.

SOLUTION From the definition on page 531 for the determinant of a 2×2 matrix, we calculate $\det A$ to be

$$\det A = a_{11}a_{22} - a_{12}a_{21}$$

$$\det A = (-1)(-3) - (2)(3) = -3$$

Since $\det A \neq 0$, the only equilibrium is the origin $\hat{\mathbf{x}} = \mathbf{0}$. This represents the situation in which the subpopulations in patches A and B are both extinct. ■

EXAMPLE 4 | Radioimmunotherapy (continued) Suppose that $a + b \neq 0$ and $c \neq 0$ in the model of Example 1. Provide a biological interpretation of these assumptions and determine the equilibrium of the model.

SOLUTION The assumptions mean that there is a nonzero loss rate of antibody from both the bloodstream ($a + b \neq 0$) and from the tumor ($c \neq 0$). We calculate $\det A$ to be

$$\det A = -(a + b)(-c) - (b)(0) = c(a + b)$$

This is nonzero given our assumptions about the constants. Therefore the only equilibrium is the origin, $\hat{\mathbf{x}} = \mathbf{0}$. This represents the situation in which there is no antibody in the bloodstream or in the tumor. ■

In Section 7.2 we introduced the concept of *local stability* in the context of single-variable differential equations. An analogous concept applies to systems of differential equations.

Definition An equilibrium $\hat{\mathbf{x}}$ of differential equation (2) is **locally stable** if $\mathbf{x}$ approaches the value $\hat{\mathbf{x}}$ as $t \rightarrow \infty$ for all initial values of $\mathbf{x}$ sufficiently close to $\hat{\mathbf{x}}$.

For systems of differential equations, stability of an equilibrium requires that all components of the vector $\mathbf{x}$ approach their equilibrium values as $t \rightarrow \infty$. As we will see in Section 10.2, for *generic* systems, if $\hat{\mathbf{x}}$ is locally stable, then it is stable for all initial conditions (not just those sufficiently close to $\hat{\mathbf{x}}$). This is referred to as **global stability**. Consequently, in this chapter we will use the terms *locally stable* and *stable* interchangeably. An equilibrium that is not stable is called **unstable**.

The remainder of this section presents three qualitatively different kinds of behavior that the system of differential equations (2) can exhibit. These are classified in terms of the nature of the equilibrium $\hat{\mathbf{x}} = \mathbf{0}$ and are called saddles, nodes, and spirals. Their qualitative features are introduced here and they are defined more precisely in Section 10.2.

■ Saddles

The equilibrium at the origin, $\hat{\mathbf{x}} = \mathbf{0}$, is called a *saddle* if, roughly speaking, the motion in the x_1x_2 -plane is toward the origin in some directions and away from it in others. As an example, consider Equation 2 with the matrix of coefficients

$$A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$$

Figure 2(a) displays the nullclines along with several solution curves in the x_1x_2 -plane. By definition, all solution curves that cross the x_1 -nullcline do so vertically because x_1 is not changing on this nullcline. Likewise, solution curves that cross the x_2 -nullcline do so horizontally. Figure 2(b) plots the components of the red solution curve in Figure 2(a) against time.

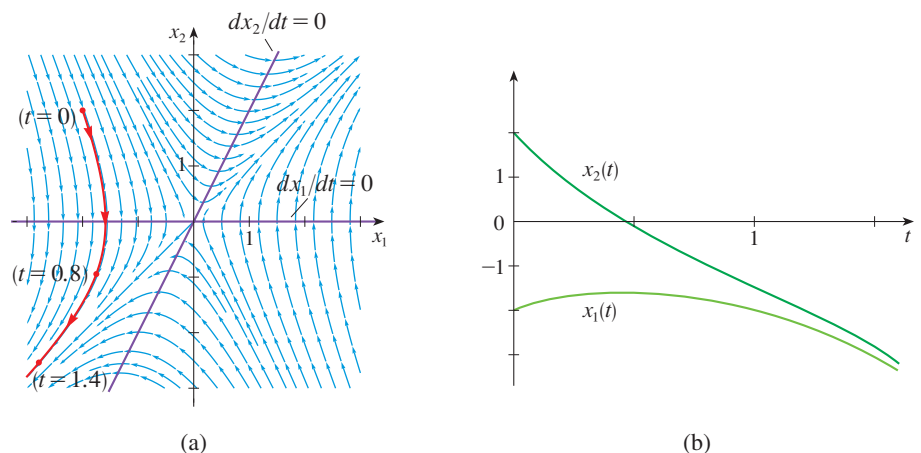


FIGURE 2

Part (a) plots the phase plane, with nullclines in purple and solution curves in blue. The red curve is a particular solution whose components are plotted against time in part (b).

When the equilibrium at the origin is a saddle, it is always unstable because there is always some initial condition for which the system moves away from the origin.

■ Nodes

The equilibrium at the origin, $\hat{\mathbf{x}} = \mathbf{0}$, is called a *node* if, roughly speaking, the motion in the phase plane is either toward the origin from all directions or away from the origin in all directions. As two examples, consider Equation 2 with the coefficients given by the matrices

$$B = \begin{bmatrix} -3 & -1 \\ -2 & -2 \end{bmatrix} \quad \text{or} \quad C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$$

Figure 3(a) displays the nullclines for matrix B , along with several solution curves of the system in the x_1x_2 -plane. Part (b) plots the components of the red solution curve in part (a) against time. Figure 4 displays the corresponding information for matrix C . Notice in Figure 3(a) and Figure 4(a) that solution curves cross the nullclines either vertically or horizontally, depending on whether the nullcline is for x_1 or x_2 , respectively.

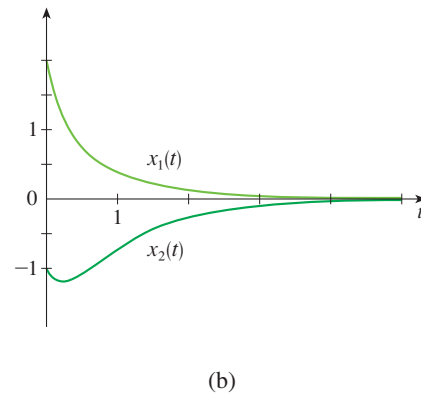
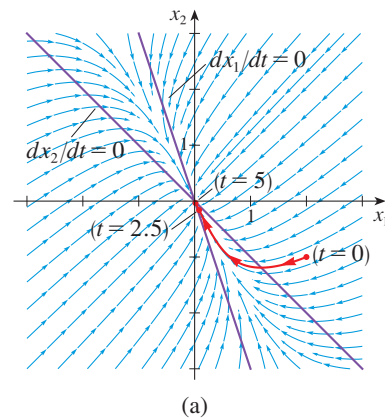


FIGURE 3

Part (a) plots the phase plane for matrix B with nullclines in purple and solution curves in blue. The red curve is a particular solution whose components are plotted against time in part (b).

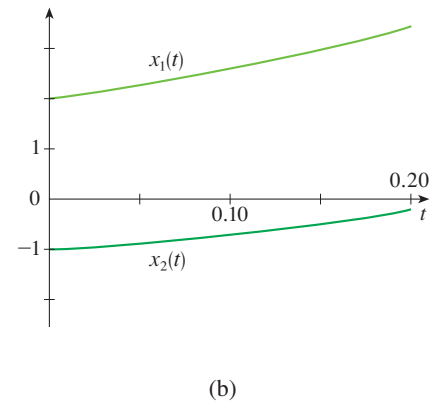
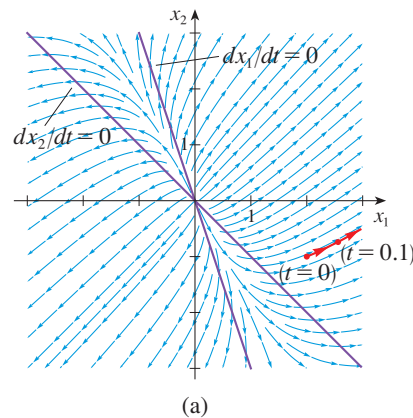


FIGURE 4

Part (a) plots the phase plane for matrix C with nullclines in purple and solution curves in blue. The red curve is a particular solution whose components are plotted against time in part (b).

The equilibrium at the origin can be a stable node, as in Figure 3(a), or an unstable node, as in Figure 4(a), depending on whether movement is toward or away from this equilibrium.

Spirals

The equilibrium at the origin, $\hat{\mathbf{x}} = \mathbf{0}$, is called a *spiral* if the motion in the phase plane is either toward the origin or away from the origin, and occurs in a spiraling fashion. As two examples, consider Equation 2 with the coefficients given by the matrices

$$D = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \text{or} \quad E = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

Figure 5 displays the nullclines for matrix D , along with several solution curves of the system in the x_1x_2 -plane. Part (b) plots the components of the red solution curve against time. Figure 6 displays the corresponding information for matrix E . Again you should make note of the orientation of solution curves where they cross a nullcline.

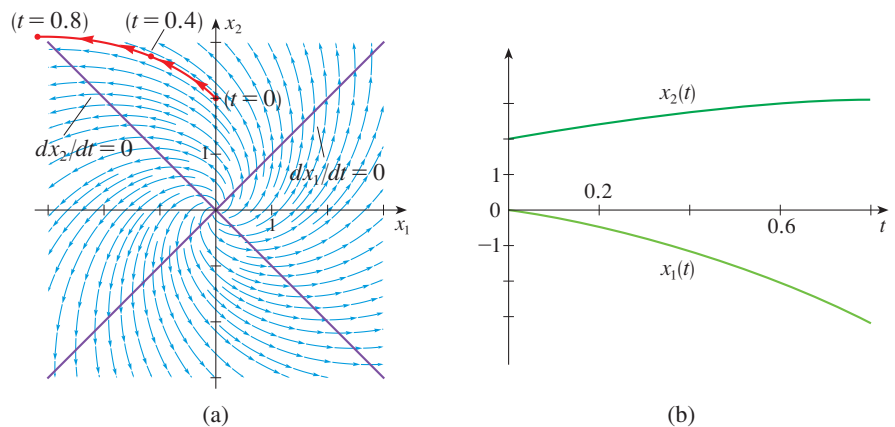


FIGURE 5
Part (a) plots the phase plane for matrix D with nullclines in purple and solution curves in blue. The red curve is a particular solution whose components are plotted against time in part (b).

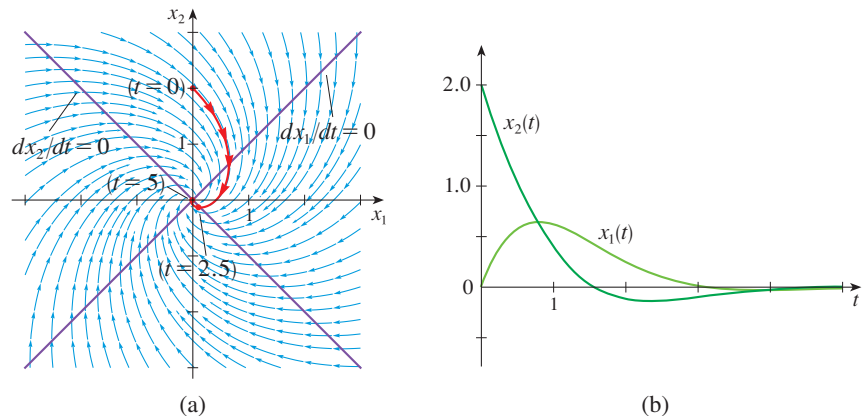


FIGURE 6
Part (a) plots the phase plane for matrix E with nullclines in purple and solution curves in blue. The red curve is a particular solution whose components are plotted against time in part (b).

The equilibrium at the origin can be an unstable spiral, as in Figure 5(a), or a stable spiral, as in Figure 6(a), depending on whether movement is away from or toward this equilibrium.

It turns out that, except for one special case called a *center*, the three qualitative types of behavior (saddle, nodes, and spirals) are the only possibilities for generic linear systems of differential equations whose matrix of coefficients is not defective. This will be demonstrated in the next section by deriving an explicit formula for the solutions of such systems.

EXERCISES 10.1

1–6 Specify whether each system is autonomous or nonautonomous, and whether it is linear or nonlinear. If it is linear, specify whether it is homogeneous or nonhomogeneous.

1. $dx/dt = x - y$, $dy/dt = -3ty + x$
2. $dy/dx = 2y$, $dz/dx = x - z + 3$
3. $dy/dt = 3yz - 2z$, $dz/dt = 2z + 5y$
4. $dy/dx = 3y - 2$, $dz/dx = 7z + y$
5. $dx/dz = 3x - 2y$, $dy/dz = 2z + 3y$
6. $dx/dt = xy - y$, $dy/dt = 4tx - xy$

7–14 Write each system of linear differential equations in matrix notation.

7. $dx/dt = 5x - 3y$, $dy/dt = 2y - x$
8. $dx/dt = x - 2$, $dy/dt = 2y + 3x - 1$
9. $dx/dt = 3ty - 7$, $dy/dt = 2x - 3y$
10. $dx/dt = 5y$, $dy/dt = 2x - y$
11. $dx/dt = 2x - 5$, $dy/dt = 3x + 7y$
12. $dx/dt = 2x - y \sin t$, $dy/dt = y - x$
13. $dx/dt = x + 4y - 3t$, $dy/dt = y - x$
14. $dx/dt = y - 2x\sqrt{t} + 7$, $dy/dt = 3x + 2$

15–22 Given the system of differential equations $dx/dt = Ax$, construct the phase plane, including the nullclines. Does the equilibrium look like a saddle, a node, or a spiral?

15. $A = \begin{bmatrix} -3 & 1 \\ 2 & -2 \end{bmatrix}$
16. $A = \begin{bmatrix} -2 & 1 \\ -1 & -1 \end{bmatrix}$
17. $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$
18. $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$
19. $A = \begin{bmatrix} -1 & 2 \\ -3 & 0 \end{bmatrix}$
20. $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
21. $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$
22. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

23. Consider the system of linear differential equations $dx/dt = Ax + \mathbf{g}$, where $\mathbf{g}$ is a vector of constants. Suppose that A is nonsingular.

- (a) What is the equilibrium of this system of equations?
- (b) Using $\hat{\mathbf{x}}$ to denote the equilibrium found in part (a), define a new vector of variables $\mathbf{y} = \mathbf{x} - \hat{\mathbf{x}}$. What do the components of $\mathbf{y}$ represent?
- (c) Show that $\mathbf{y}$ satisfies the differential equation $dy/dt = A\mathbf{y}$. This demonstrates how we can reduce a nonhomogeneous system of linear differential equations

to a system that is homogeneous by using a change of variables.

24. Consider the system of linear differential equations

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix} \mathbf{x}$$

The system is *nongeneric*, that is, the determinant of the matrix of coefficients is zero.

- (a) There are an infinite number of equilibria, all lying on a line in the phase plane. What is the equation of this line?
- (b) Construct the phase plane for this system.

25. Consider an autonomous homogeneous system of linear differential equations with coefficient matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Suppose that $\det A = 0$. Show that there are an infinite number of equilibria.

26. Consider the following homogeneous system of three linear differential equations:

$$\begin{aligned} dx/dt &= 3x + 2y - z \\ dy/dt &= x - y - z \\ dz/dt &= y + 3z \end{aligned}$$

Suppose that $x + y = 5$ at all times. Show that this system can be reduced to two nonhomogeneous linear differential equations given by

$$\begin{aligned} dx/dt &= x - z + 10 \\ dz/dt &= -x + 3z + 5 \end{aligned}$$

27. Consider the following homogeneous system of four linear differential equations:

$$\begin{aligned} dw/dt &= 2x + y - z \\ dx/dt &= 3x + z \\ dy/dt &= -y + 2z \\ dz/dt &= 3x - 5y \end{aligned}$$

Suppose that $x + z = 2$ and $y + w = 3$ at all times. Show that this system can be reduced to two nonhomogeneous linear differential equations given by

$$\begin{aligned} dw/dt &= 3x - w + 1 \\ dx/dt &= 2x + 2 \end{aligned}$$

28. Consider any homogeneous, autonomous system of three linear differential equations for which the variables satisfy $ax_1 + bx_2 + cx_3 = d$, where a, b, c , and d are constants, not all of which are zero. Show that the system can be reduced to a nonhomogeneous system of two linear differential equations.

29. Second-order linear differential equations take the form

$$y''(t) + p(t)y'(t) + q(t)y(t) = g(t)$$

where p , q , and g are continuous functions of t . Suppose we have initial conditions $y(0) = a$ and $y'(0) = b$. Show that this equation can be rewritten as a system of two first-order linear differential equations having the form

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 1 \\ -q(t) & -p(t) \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ g(t) \end{bmatrix}$$

with $\mathbf{x}(0) = \begin{bmatrix} a \\ b \end{bmatrix}$

where $x_1(t) = y(t)$ and $x_2(t) = y'(t)$.

30. **Metapopulation dynamics** Example 2 presents a model for a population of deer mice that is split into two patches through habitat fragmentation. The model is

$$\frac{dx_A}{dt} = -x_A + 2x_B \quad \frac{dx_B}{dt} = 3x_A - 3x_B$$

where x_A and x_B are the population sizes in patches A and B, respectively.

- Construct the phase plane, including the nullclines.
- Describe what happens to the population in each patch as $t \rightarrow \infty$ if both start with nonzero sizes.

31. **Gene regulation** Genes produce molecules called mRNA that then produce proteins. High levels of protein can inhibit the production of mRNA, resulting in a feedback that regulates gene expression. Using m and p to denote the amounts of mRNA and protein in a cell ($\times 10^2$ copies/cell), a simple model of gene regulation is

$$\begin{aligned} dm/dt &= 1 - p - m \\ dp/dt &= m - p \end{aligned}$$

Construct the phase plane, including the nullclines.
[Hint: This system is nonhomogeneous.]

32. **Prostate cancer** During treatment, tumor cells in the prostate can become resistant through a variety of biochemical mechanisms. Some of these are reversible—the cells revert to being sensitive once treatment stops—and some are not. Using x_1 , x_2 , and x_3 to denote the fraction of cells that are sensitive, temporarily resistant, and permanently resistant, respectively, a simple model for their dynamics during treatment is

$$\begin{aligned} dx_1/dt &= -ax_1 - cx_1 + bx_2 \\ dx_2/dt &= ax_1 - bx_2 - dx_2 \\ dx_3/dt &= cx_1 + dx_2 \end{aligned}$$

Use the fact that $x_1 + x_2 + x_3 = 1$ to reduce this to a nonhomogeneous system of two linear differential equations for x_1 and x_3 .

Source: Adapted from Y. Hirata et al., "Development of a Mathematical Model that Predicts the Outcome of Hormone Therapy for Prostate Cancer," *Journal of Theoretical Biology* 264 (2010): 517–27.

33. **Radioimmunotherapy** Example 1 presents a model of radioimmunotherapy. The model is

$$\frac{dx_1}{dt} = -ax_1 - bx_1 \quad \frac{dx_2}{dt} = bx_1 - cx_2$$

where x_1 and x_2 denote the amount of antibody (in μg) in the bloodstream and tumor, respectively, at time t (in minutes), and all constants are positive.

- Construct the phase plane, including the nullclines, to determine the qualitative behavior of the system.
- Describe what happens to the amount of antibody in each part of the body as $t \rightarrow \infty$.

Source: Adapted from A. Flynn et al., "Effectiveness of Radiolabelled Antibodies for Radio-Immunotherapy in a Colorectal Xenograft Model: A Comparative Study using the Linear-Quadratic Formulation," *International Journal of Radiation Biology* 77 (2001): 507–17.

34. **Jellyfish locomotion** Jellyfish move by contracting an elastic part of their body, called a bell, that creates a high-pressure jet of water. When the contractive force stops, the bell then springs back to its natural shape. Jellyfish locomotion has been modeled using a second-order linear differential equation having the form

$$mx''(t) + bx'(t) + kx(t) = 0$$

where $x(t)$ is the displacement of the bell at time t , m is the mass of the bell (in grams), b is a measure of the friction between the bell and the water (in units of $\text{N/m} \cdot \text{s}$), and k is a measure of the stiffness of the bell (in units of N/m). Suppose that $m = 100$ g, $b = 0.1$ $\text{N/m} \cdot \text{s}$, and $k = 1$ N/m .

- Define the new variables $z_1(t) = x(t)$ and $z_2(t) = x'(t)$, and show that the model can be expressed as a system of two first-order linear differential equations.
- Construct the phase plane, including the nullclines, for the equations from part (a).

Source: Adapted from M. DeMont et al., "Mechanics of Jet Propulsion in the Hydromedusan Jellyfish, *Polyorchis Penicillatus*: III. A Natural Resonating Bell—The Presence and Importance of a Resonant Phenomenon in the Locomotor Structure," *Journal of Experimental Biology* 134 (1988): 347–61.



10.2 Solving Systems of Linear Differential Equations

Most systems of linear differential equations that arise in the life sciences take the form of an *initial-value problem*. Suppose A is a 2×2 matrix of coefficients and $\mathbf{x}$ is a two-dimensional vector whose components are variables.

(1) Definition The system of linear differential equations

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

together with an initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ is called an **initial-value problem**.

The initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ specifies the values that each of the component variables in $\mathbf{x}(t)$ must take at $t = t_0$.

A solution to the initial-value problem (1) is a vector $\mathbf{x}(t)$ whose two components are each functions of t and that together satisfy the differential equation and initial condition. More specifically, we say that a function $\boldsymbol{\varphi}(t)$ is a solution of the differential equation in (1) if, when substituted into the left and right sides of the differential equation, it produces an equality. Further, if $\boldsymbol{\varphi}(t_0) = \mathbf{x}_0$ as well, then we say $\boldsymbol{\varphi}(t)$ solves the initial-value problem (1).

Just as there were many solutions to the differential equations we examined in Chapter 7, there are typically many solutions to a system of linear differential equations. Only one of these, however, will satisfy the initial condition. The following result can be proved.

(2) Theorem: Existence and Uniqueness of Solutions The initial-value problem (1) has one and only one solution $\mathbf{x}(t)$, and this solution is defined for all $t \in \mathbb{R}$.

Theorem 2 tells us two things: First, it tells us that there will always be a solution to the initial-value problem. Second, it tells us that there will be only one solution. Therefore, if we can manage to find a solution, no matter how we do it, then we are guaranteed that it will be the only one.

How can we find a solution to the initial-value problem? Consider a simple example:

$$(3) \quad \frac{dx_1}{dt} = -x_1 \quad \frac{dx_2}{dt} = -3x_2$$

with initial condition $x_1(0) = 2$ and $x_2(0) = 5$. In matrix form this is

$$(4) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

where $A = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}$ and $\mathbf{x}(0) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

In Equations 3 the differential equation for x_1 does not involve x_2 , and vice versa. Their solutions are therefore readily found to be

$$x_1(t) = c_1 e^{-t} \quad x_2(t) = c_2 e^{-3t}$$

where c_1 and c_2 are constants to be determined from the initial conditions (for example, see Section 3.6). Writing these solutions in vector form, we have

$$(5) \quad \mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} c_1 e^{-t} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ c_2 e^{-3t} \end{bmatrix} = c_1 e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Solution (5) is composed of the sum of two parts, each being an exponential function multiplied by a vector. Defining

$$\mathbf{x}_1(t) = e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_2(t) = e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

we can write solution (5) as

$$\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$$

In fact, $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are each, individually, solutions to the system of differential equations (4). To see this, substitute $\mathbf{x}_1(t)$ into the left and right sides of Equation 4. The left side gives

$$\frac{d\mathbf{x}_1}{dt} = -e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and the right side gives

$$Ae^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = e^{-t} \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = e^{-t} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The two sides are identical, which means that $\mathbf{x}_1(t)$ is a solution. It is plotted in the phase plane in Figure 1(a). Likewise, substituting $\mathbf{x}_2(t)$ into the left side results in

$$\frac{d\mathbf{x}_2}{dt} = -3e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and the right side gives

$$Ae^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e^{-3t} \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e^{-3t} \begin{bmatrix} 0 \\ -3 \end{bmatrix} = -3e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Again these sides are identical and so $\mathbf{x}_2(t)$ is a solution as well. It is plotted in Figure 1(b).

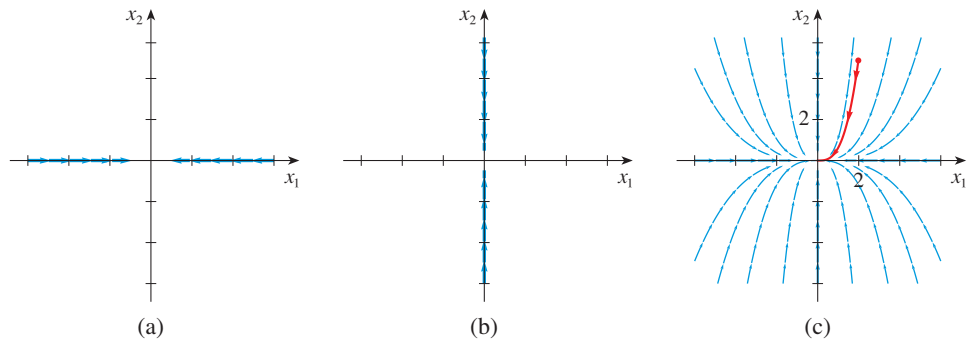


FIGURE 1 Blue curves are solutions to the system of differential equations. Part (a) shows solutions $c_1 \mathbf{x}_1(t)$ for different values of c_1 . Part (b) shows solutions $c_2 \mathbf{x}_2(t)$ for different values of c_2 . Part (c) shows solutions $c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$ for different values of c_1 and c_2 . The red curve is the solution to the initial-value problem.

Equation 5 is a family of solutions to the system of differential equations (4). Theorem 2 tells us, however, that there is a unique solution to this system of equations once we take the initial condition into account. If solution (5) is to satisfy the initial condition, then we require that

$$\mathbf{x}(0) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Therefore we must choose $c_1 = 2$ and $c_2 = 5$, and the unique solution to the initial-value problem (4) is

$$\mathbf{x}(t) = 2e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

This solution is plotted in the phase plane in Figure 1(c).

■ The General Solution

The preceding calculations illustrate some general principles that apply to the initial-value problem (1): First, as we will see, solutions to the differential equation in (1) are always functions of the form $e^{\lambda t}\mathbf{v}$, where λ is a scalar and $\mathbf{v}$ is a vector. Second, the sum of any two solutions to the differential equation is itself a solution. This latter fact is summarized as follows. (See also Exercise 9.)

(6) The Superposition Principle Suppose $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions of the system of differential equations in (1). Then any function of the form $\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$, where c_1 and c_2 are scalar quantities, is also a solution.

Typically, a family of solutions to the differential equation in (1) is obtained by finding the values of λ and $\mathbf{v}$ for which $e^{\lambda t}\mathbf{v}$ is a solution, and then combining these solutions using the Superposition Principle (6). The unique solution to the initial-value problem is then obtained by choosing appropriate constants c_1 and c_2 .

Let's put these ideas into practice for the initial-value problem (1) with

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

We begin by looking for solutions having the form $e^{\lambda t}\mathbf{v}$ for some λ and $\mathbf{v}$. Substituting this function into the differential equation in (1) gives

$$\frac{d}{dt}(e^{\lambda t}\mathbf{v}) = Ae^{\lambda t}\mathbf{v}$$

or

$$\lambda e^{\lambda t}\mathbf{v} = Ae^{\lambda t}\mathbf{v}$$

Because $e^{\lambda t}$ is never zero, we can divide both sides by $e^{\lambda t}$, giving

$$(7) \quad \lambda \mathbf{v} = A\mathbf{v}$$

From Definition 8.7.2 we see that the values of $\mathbf{v}$ and λ that satisfy Equation 7 are the eigenvector-eigenvalue pairs of the matrix A . This shows that a function of the form $e^{\lambda t}\mathbf{v}$ is a solution to the differential equation in (1) provided that $\mathbf{v}$ and λ are an eigenvector-eigenvalue pair of A .

Our focus in the remainder of this chapter will be on coefficient matrices that are not defective and therefore that have distinct eigenvalues (advanced courses in differential equations treat the general case). Recall from page 538 that such matrices are guaranteed

to have distinct eigenvector-eigenvalue pairs. Denoting these by $\mathbf{v}_i$ and λ_i , we can then use the Superposition Principle (6) to obtain the family of solutions

$$(8) \quad \mathbf{x}(t) = c_1 e^{\lambda_1 t} \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + c_2 e^{\lambda_2 t} \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix}$$

where $\begin{bmatrix} | \\ \mathbf{v}_i \\ | \end{bmatrix}$ is the i th eigenvector. Equation 8 is called the **general solution** of the differential equation in (1). It is valid for any two-dimensional system whose coefficient matrix has distinct eigenvalues (even the nongeneric case where $\det A = 0$).

Finally, we can use the initial condition to find the values of c_1 and c_2 that provide the unique solution to the initial-value problem guaranteed by Theorem 2. We have

$$\mathbf{x}(0) = c_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + c_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

or

$$(9) \quad \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

Since the eigenvalues of A are distinct, a theorem from linear algebra says that the matrix whose columns are the associated eigenvectors is nonsingular (see Exercise 10). Therefore Theorem 8.6.4 tells us that Equation 9 has a unique solution for c_1 and c_2 .

Nullclines versus Eigenvectors

The general solution (8) shows that the eigenvectors of the matrix A in the initial-value problem (1) play an important role in the behavior of the variable $\mathbf{x}(t)$. Section 10.1 revealed that the nullclines of the differential equation in (1) also play an important role. What is the relationship between the two?

Recall that the nullclines for two-variable systems of linear differential equations are straight lines in the plane along which the rate of change of one of the variables is zero (see Section 10.1). This means that the motion in the plane is either vertical or horizontal along nullclines [see Figure 2(a)].

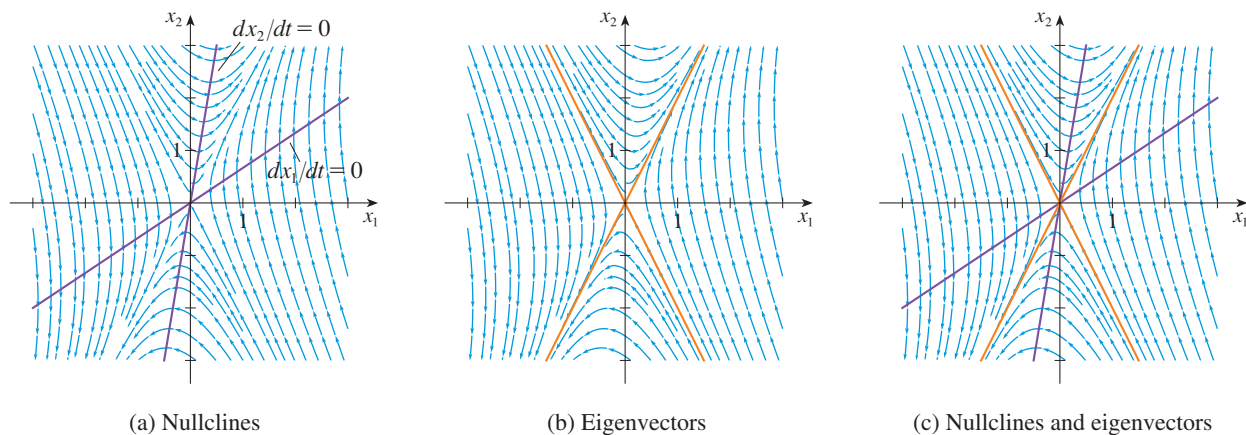


FIGURE 2 Purple lines are nullclines. Orange lines indicate all scalar multiples of the eigenvectors. Blue curves are solutions.

The eigenvectors of the matrix A appear in the general solution (8) in two terms, each having the form

$$(10) \quad ce^{\lambda t} \mathbf{v}$$

and each of these two terms is itself a solution to differential equation (1). What is the geometric interpretation of (10)? The factor c is a scalar, as is the factor $e^{\lambda t}$. Therefore $ce^{\lambda t}$ is a scalar quantity and it varies in magnitude with t . The remaining factor $\mathbf{v}$ is the eigenvector associated with λ . Thus (10) is a scalar multiple of the vector $\mathbf{v}$, where $ce^{\lambda t}$ is the scalar. Section 8.2 demonstrated that, geometrically, scalar multiplication simply scales the length of a vector. Therefore the plot of (10) in the phase plane traces out a straight line, moving in the direction of eigenvector $\mathbf{v}$ as t increases.

Exercise 39 demonstrates that solution curves in the phase plane cannot cross. Since (10) defines a solution curve that is a scalar multiple of the eigenvector $\mathbf{v}$, the lines corresponding to all scalar multiples of the eigenvectors therefore constrain the direction of the solution curves in the phase plane [see Figure 2(b)]. Figure 2(c) shows how, together, the nullclines and the eigenvectors dictate the shape of the solution curves in the phase plane.

We now return to the three qualitative kinds of behavior documented in Section 10.1: saddles, nodes, and spirals. Throughout the remainder of this section we assume that $\det A \neq 0$ and therefore the origin is the only equilibrium. We also assume that the eigenvalues of A are distinct.

■ Saddles

The origin is called a **saddle** if the eigenvalues of the coefficient matrix are real and have opposite signs. The form of solution (8) immediately shows that the origin is unstable in this case because one of the exponential terms will always increase without bound. As a result, there is always some initial condition for which $\mathbf{x}(t)$ does not approach the origin as $t \rightarrow \infty$.

The saddle explored on page 635 used the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$$

The eigenvalues of A are $\lambda_1 = -2$ and $\lambda_2 = 1$. The associated eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The general solution (8) is therefore

$$\mathbf{x}(t) = c_1 e^{-2t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + c_2 e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

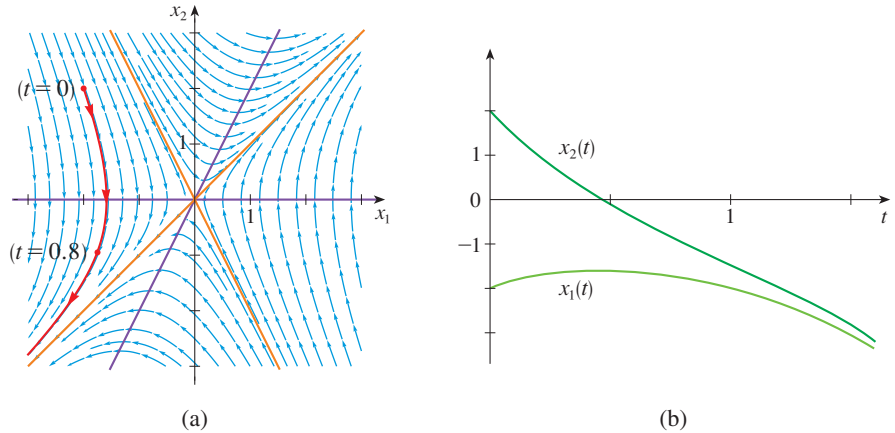
To obtain a unique solution we need to specify an initial condition. Suppose that $\mathbf{x}(0) = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$. Equation 9 then becomes

$$\begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

The solution to this equation is $c_1 = \frac{4}{3}$ and $c_2 = -\frac{2}{3}$. Figure 3 displays several solution curves, along with the nullclines, the eigenvectors, and the solution to the initial-value problem.

FIGURE 3

Part (a) plots the phase plane, with nullclines in purple, all scalar multiples of the eigenvectors in orange, and solution curves in blue. The red curve is the solution to the initial-value problem, and its components are plotted against time in part (b).



■ Nodes

The origin is called a **node** if the eigenvalues of A are real and have the same sign. The form of solution (8) reveals that the origin is unstable if the eigenvalues are positive (an *unstable node*) and stable if the eigenvalues are negative (a *stable node*).

For the nodes explored on page 636 the coefficient matrices were

$$B = \begin{bmatrix} -3 & -1 \\ -2 & -2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$$

The eigenvalues of B are $\lambda_1 = -1$ and $\lambda_2 = -4$, with eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The eigenvalues of C are $\lambda_1 = 1$ and $\lambda_2 = 4$, with eigenvectors again being

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The general solutions are therefore

$$\mathbf{x}(t) = c_1 e^{-t} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + c_2 e^{-4t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for matrix } B$$

$$\text{and} \quad \mathbf{x}(t) = c_1 e^t \begin{bmatrix} -1 \\ 2 \end{bmatrix} + c_2 e^{4t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for matrix } C$$

To obtain unique solutions in each case we need to specify an initial condition. Suppose that, in both cases,

$$\mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Equation 9 then becomes

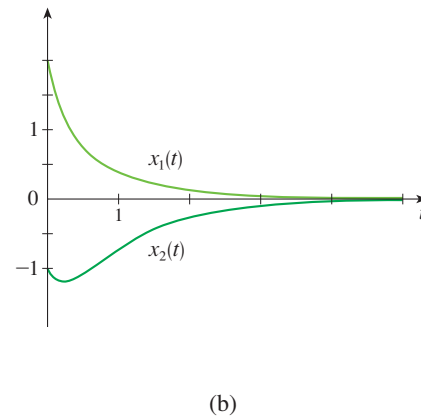
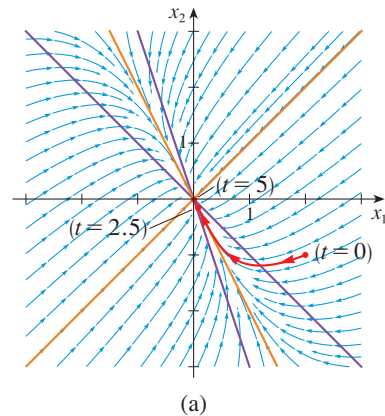
$$\begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

in both cases. The solution to this equation is $c_1 = -1$ and $c_2 = 1$. Figures 4 and 5 dis-

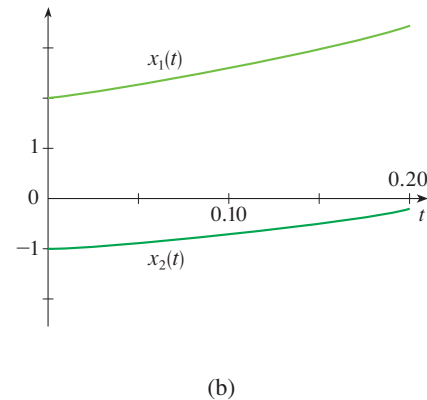
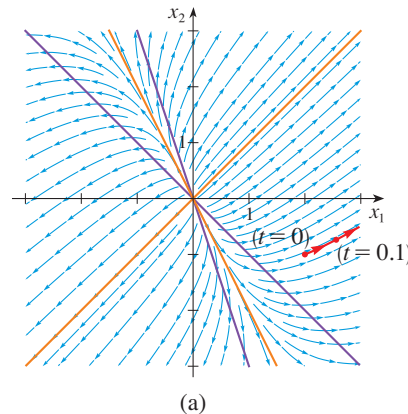
play several solution curves, along with the nullclines, the eigenvectors, and the solutions to each of these initial-value problems.

FIGURE 4

Part (a) plots the phase plane for matrix B , with nullclines in purple, all scalar multiples of the eigenvectors in orange, and solution curves in blue. The red curve is the solution to the initial-value problem, and its components are plotted against time in part (b).

**FIGURE 5**

Part (a) plots the phase plane for matrix C , with nullclines in purple, all scalar multiples of the eigenvectors in orange, and solution curves in blue. The red curve is the solution to the initial-value problem, and its components are plotted against time in part (b).



■ Spirals

See Appendix G for a review of complex numbers.

The remaining possibility is that the eigenvalues of the coefficient matrix are complex conjugates, taking the form $\lambda = a \pm bi$ for some real numbers a and b . In this case the origin is a spiral.

To see how complex eigenvalues lead to such behavior, we first note that the eigenvectors associated with complex conjugate eigenvalues are themselves complex conjugates (see Exercises 8.8.26 and 8.8.27). Therefore the eigenvectors associated with the eigenvalues $\lambda = a \pm bi$ can be written $\mathbf{v} = \mathbf{u} \pm i\mathbf{w}$, where $\mathbf{u}$ and $\mathbf{w}$ are real-valued vectors. We take each eigenvector-eigenvalue pair in turn.

To evaluate the first term of solution (8) we substitute $\lambda = a + ib$ and $\mathbf{v} = \mathbf{u} + i\mathbf{w}$ into $e^{\lambda t}\mathbf{v}$ to obtain

$$(11) \quad e^{(a+bi)t}(\mathbf{u} + i\mathbf{w})$$

Euler's formula

$$e^{iy} = \cos y + i \sin y$$

is discussed in Appendix G.

Euler's formula allows us to simplify this further. If $x + iy$ is a complex number, then e^{x+iy} can be rewritten as

$$\begin{aligned} e^{x+iy} &= e^x e^{iy} \\ &= e^x (\cos y + i \sin y) \end{aligned}$$

Applying this result with $x = at$ and $y = bt$, we can write (11) as

$$\begin{aligned} e^{(a+bi)t}(\mathbf{u} + i\mathbf{w}) &= e^{at}(\cos bt + i \sin bt)(\mathbf{u} + i\mathbf{w}) \\ &= e^{at}(\mathbf{u} \cos bt - \mathbf{w} \sin bt + i\mathbf{w} \cos bt + i\mathbf{u} \sin bt) \\ &= \mathbf{g}(t) + i\mathbf{h}(t) \end{aligned}$$

where, to simplify notation, we have defined

$$\mathbf{g}(t) = e^{at}(\mathbf{u} \cos bt - \mathbf{w} \sin bt)$$

and

$$\mathbf{h}(t) = e^{at}(\mathbf{w} \cos bt + \mathbf{u} \sin bt)$$

To calculate the second term of solution (8) we can follow the same steps for the complex conjugate eigenvector-eigenvalue pair, obtaining

$$e^{(a-bi)t}(\mathbf{u} - i\mathbf{w}) = \mathbf{g}(t) - i\mathbf{h}(t)$$

Therefore solution (8) can be written as

$$(12) \quad \mathbf{x}(t) = c_1[\mathbf{g}(t) + i\mathbf{h}(t)] + c_2[\mathbf{g}(t) - i\mathbf{h}(t)]$$

Although (12) is the general solution in the case of complex eigenvalues, this family of solutions is complex-valued and we are interested only in real-valued solutions. We can restrict our attention to a family of real-valued solutions by choosing the constants $c_1 = (k_1 - ik_2)/2$ and $c_2 = (k_1 + ik_2)/2$, where k_1 and k_2 are arbitrary, real constants. Although we have not formally justified the use of complex values of c_1 and c_2 in solution (8), this does, in fact, turn out to be valid. With these choices, the imaginary part of Equation 12 then disappears:

$$\begin{aligned} \mathbf{x}(t) &= \frac{k_1 - ik_2}{2} [\mathbf{g}(t) + i\mathbf{h}(t)] + \frac{k_1 + ik_2}{2} [\mathbf{g}(t) - i\mathbf{h}(t)] \\ &= \frac{1}{2} [2k_1\mathbf{g}(t) + ik_1\mathbf{h}(t) - ik_1\mathbf{h}(t) - ik_2\mathbf{g}(t) + ik_2\mathbf{g}(t) + 2k_2\mathbf{h}(t)] \\ &= k_1\mathbf{g}(t) + k_2\mathbf{h}(t) \end{aligned}$$

Therefore, from the definitions of $\mathbf{g}(t)$ and $\mathbf{h}(t)$, the family of real solutions is

$$(13) \quad \mathbf{x}(t) = k_1 e^{at}(\mathbf{u} \cos bt - \mathbf{w} \sin bt) + k_2 e^{at}(\mathbf{w} \cos bt + \mathbf{u} \sin bt)$$

Equation 13 is the (real-valued) general solution when the eigenvalues are complex.

To obtain the unique solution to the initial-value problem, we need to make use of an initial condition. In particular, supposing that

$$\mathbf{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

then, from Equation 13, we have

$$\mathbf{x}(0) = k_1\mathbf{u} + k_2\mathbf{w}$$

or

$$(14) \quad \begin{bmatrix} | & | \\ \mathbf{u} & \mathbf{w} \\ | & | \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

Although we won't prove it, a matrix whose columns are $\mathbf{u}$ and $\mathbf{w}$ (that is, whose columns are the real and imaginary parts of the eigenvectors) is nonsingular. Therefore Theorem 8.6.4 says that Equation 14 has a unique solution for k_1 and k_2 . This then gives the unique solution to the initial-value problem.

The form of Equation 13 also reveals that the origin will be an *unstable spiral* if the real part a of the eigenvalues is positive, and a *stable spiral* if the real part a is negative. The case where $a = 0$ is called a **center** and the solutions in this case form closed curves in the phase plane (see Exercise 30).

Let's consider the two coefficient matrices we explored in Section 10.1:

$$D = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad E = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

The eigenvalues of D are $\lambda = 1 \pm i$ with eigenvectors

$$\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \pm i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Therefore

$$a = 1, \quad b = 1, \quad \mathbf{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Equation 13 then simplifies to

$$\begin{aligned} \mathbf{x}(t) &= k_1 e^t \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cos t - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \sin t \right) + k_2 e^t \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin t \right) \\ &= e^t \begin{bmatrix} -k_1 \sin t + k_2 \cos t \\ k_1 \cos t + k_2 \sin t \end{bmatrix} \end{aligned}$$

The eigenvalues of E are $\lambda = -1 \pm i$, with eigenvectors

$$\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \pm i \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Therefore

$$a = -1, \quad b = 1, \quad \mathbf{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Equation 13 in this case becomes

$$\mathbf{x}(t) = e^{-t} \begin{bmatrix} k_1 \sin t - k_2 \cos t \\ k_1 \cos t + k_2 \sin t \end{bmatrix}$$

Finally, to obtain unique solutions in each case we need to specify an initial condition. Suppose that, in both cases,

$$\mathbf{x}(0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Equation 14 then becomes

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \text{for matrix } D$$

$$\text{or} \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \text{for matrix } E$$

In both cases the solution is $k_1 = 2$ and $k_2 = 0$. Figures 6 and 7 display several solution curves, along with the nullclines and the solutions to each of these initial-value prob-

lems. Note that the eigenvectors cannot be plotted in the phase plane because they are complex-valued.

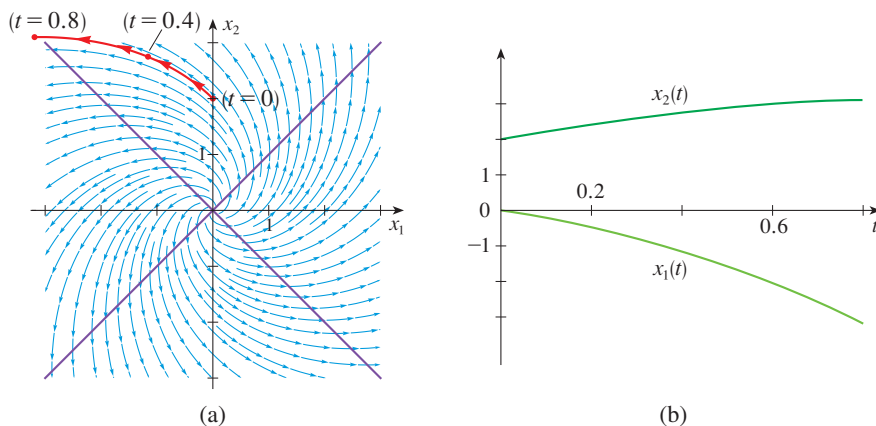


FIGURE 6

Part (a) plots the phase plane for matrix D , with nullclines in purple and solution curves in blue. The red curve is the solution to the initial-value problem, and its components are plotted against time in part (b).

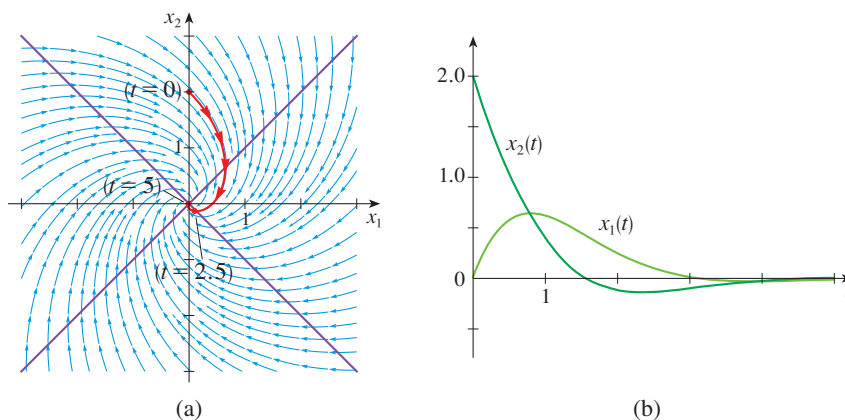


FIGURE 7

Part (a) plots the phase plane for matrix E , with nullclines in purple and solution curves in blue. The red curve is the solution to the initial-value problem, and its components are plotted against time in part (b).

■ Long-Term Behavior

Equation 8 for the solution to two-dimensional systems can be used to prove the following theorem about long-term behavior.

(15) Theorem The origin of the initial-value problem (1) is a stable equilibrium if and only if the real parts of both eigenvalues of A are negative.

Although we have focused only on the case where the coefficient matrix has distinct eigenvalues, Theorem 15 holds in general.

Another useful result relates the determinant and the trace of A to the stability of the origin (recall that the trace of a matrix is the sum of its diagonal elements).

(16) Theorem The origin of initial-value problem (1) is a stable equilibrium if and only if $\det A > 0$ and $\text{trace } A < 0$.

Theorems 15 and 16 are proved in Exercises 36 and 37. The stability properties of the equilibrium at the origin can be summarized graphically in terms of the determinant and